

The Browser Landscape 2026

Features, categories & UI/UX structure of every major browser
+ the Andromeda v2.0 gap analysis

Andromeda Project — v3 Discovery Research
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The Browser Landscape 2026 — Full Research Report

Project: Andromeda (v3 discovery) **Date:** 2026-07-12 **Purpose:** Owner-requested research before v3 scoping: catalogue what every major browser does, how features are categorized and placed in the UI, and measure exactly where Andromeda v2.0 stands. Detailed per-browser dossiers live in [browsers/](#) — this document is the synthesis.

Browsers covered: Chrome, Firefox, Edge, Samsung Internet, Brave, Opera/Opera GX, Vivaldi, DuckDuckGo, Kiwi (legacy), the AI-first wave (Comet, Arc Search, Dia, Atlas), plus the desktop feature superset (Chrome/Firefox/Edge desktop) and a dedicated cross-browser UI/UX structure study.

1. Executive summary

- The market is a pyramid.** Chrome (~65% mobile share) wins on defaults + Google integration, not feature count. Samsung Internet (~2.8%) is the one-handed UX benchmark. Brave is the ad-blocking benchmark. Firefox is the extensions/privacy benchmark. Opera is the "everything built-in" benchmark. Every successful non-Chrome browser survives by being *the best in the world at one thing* — nobody beats Chrome at being Chrome.
- 2026's battleground is AI** — but the evidence says summarize/ask-the-page are the daily-driver winners, while agentic "browse for me" remains fragile and has unsolved prompt-injection security problems. OpenAI killed its own browser (Atlas) after 9 months.
- The UI consensus has settled:** bottom-anchored unified toolbar, grid tab switcher with a private pane, icon-row + short-list menus, editable shortcut-grid home pages (feeds are the most complained-about feature in the industry), quieter permission chips, and gestures as accelerators-not-requirements. **Andromeda v2's Orbit design already matches most of this consensus** — bottom Command Bar, thumbnail grid switcher, no feed. Our structure is modern; our *depth* is not.
- WebView draws a hard line.** Extensions, engine-level fingerprint defense, DNS-over-HTTPS choice, and uBlock-parity blocking are engine-only — permanently out of reach without a Chromium fork. But the research also identified the open lanes that are 100% buildable on WebView and under-served industry-wide: **multi-profile/containers, desktop-grade tab management, a real download engine, reading list + offline, per-site settings memory, omnibox power features, and assistant-tier AI.**
- Measured honestly against the ~200-feature universe in this report, Andromeda v2 covers roughly 25–30% of what a 2026 daily driver ships** — better than the owner's felt "5%," but the missing 70% includes the features users touch most (passwords/autofill, translate, tab groups, sync, PDF viewer, TTS). Section 6 maps every gap.

2. The feature taxonomy — what a browser IS in 2026

Synthesized across all ten dossiers, every browser feature falls into 16 categories. This is the master checklist for any roadmap:

#	Category	Table-stakes examples	Benchmark browser
1	Navigation & address bar	Combined omnibox, suggestions, voice, QR scan, position choice	Chrome (omnibox intelligence)
2	Tabs & tab management	Grid switcher, groups, search, recently-closed, inactive-tab archiving, pinning	Opera (3 layouts, Islands, 100-undo) / Chrome (sync + declutter)
3	Home / new-tab page	Editable shortcut grid, optional (off-by-default) feed, wallpapers	Brave (privacy stats) / Opera (speed dial)
4	Search & discovery	Engine choice, custom engines, suggestions, visual/voice search	Chrome (Lens) / Brave (own index)
5	Privacy & security	Tracker blocking, private mode + biometric lock, HTTPS-only, Safe Browsing, cookie controls, permission management	Firefox (ETP/TCP) / Brave (farbling)
6	Ad/content blocking	Filter lists, cosmetic filtering, per-site toggle, element zapping, custom lists	Brave (Rust engine, uBO lists, scriptlets)
7	Passwords & autofill	Password manager, generator, breach alerts, passkeys, address/payment autofill	Chrome (GPM) / Samsung (Knox vault)
8	Downloads & media	Pause/resume, parallel, background video, PiP, video assistant, offline pages	Samsung (Video Assistant) / Brave (background YouTube)
9	Reading & content	Reader mode, reading list, full-page translate, TTS/read-aloud, PDF viewer, find-in-page	Edge (Immersive Reader + Read Aloud + Translate)
10	Customization & appearance	Theme, bar position, toolbar/menu customization, force-dark, wallpapers	Samsung (drag-and-drop 7-slot toolbar + menu)
11	Sync & account	Bookmarks/history/tabs/passwords sync, E2E encryption, send-to-device, no-account pairing	Brave/DDG (accountless QR chains) / Firefox (E2E by design)

#	Category	Table-stakes examples	Benchmark browser
12	AI features	Summarize, ask-the-page, AI omnibox, cross-tab reasoning, voice; (agentic = frontier)	Comet (reference) / Edge (deepest incumbent)
13	Gestures & shortcuts	Bar-swipe tab switch, pull-to-refresh, long-press menus, app shortcuts	Opera GX (FAB + haptics)
14	Performance & data	Preloading, tab sleeping, data saver, battery-friendly dark	Opera Mini (compression)
15	Extensions / scripting	Full extensions (engine-only), content-blocker plugins, user scripts	Firefox (open ecosystem) —  engine-bound
16	Accessibility & extras	Text scaling, force-zoom, high contrast, TalkBack, error-page delight, PWA install	Edge (a11y) / Chrome (dino, WebAPK)

3. UI/UX structure — the industry consensus (and where features live)

The full study is in [browsers/09-uiux-patterns.md](#). The 15 evidence-backed rules, condensed:

Layout 1. Unified address bar + toolbar at the **bottom** (thumb-zone research; Safari/Arc/Comet default bottom; Chrome/Edge/Firefox/Samsung all offer it). Dock it — never float. Never split bar and toolbar across edges. (*Andromeda:  already bottom-docked.*) 2. Persistent toolbar = address field + max 4 buttons (tabs, menu, + up to two more), ideally with **one user-configurable slot** (the Chrome/Vivaldi/Firefox compromise; Samsung's full drag-and-drop reorder is the power-user gold standard).

Tabs 3. 2-column grid of preview cards, list-view option, swipe-to-close, segmented private pane, **theme-shift for private mode** so the mode is legible at a glance. 4. Tab count badge, two characters max — Chrome shows ":D" at 100+ (delight from constraint).

Menus — the three-way division of labor every browser obeys: 5. **Menu** = weekly actions (new tab, history, downloads, bookmarks, share, find, translate, desktop site). Consensus shape: icon action-row on top + short grouped list below (Chrome M3 and Firefox 141 converged independently). Samsung/Edge use customizable icon grids instead. 6. **Settings** = rarely-visited preferences, 4 shallow groups: Account → Basics (search engine, bar position, theme) → Privacy & security → Advanced. Settings search box appears once settings get big (Samsung v27). 7. **Long-press** = ALL contextual actions (links, images, selection, address bar, home tiles). Never in the main menu.

Home page 8. Search box + user-editable shortcut grid. **No feed by default** — content feeds are the most complained-about browser feature industry-wide (Edge MSN, Chrome Discover, Firefox Pocket). User-editable = accepted; algorithm-pushed = resented.

Gestures (standard vocabulary users already know) 9. Pull-to-refresh · horizontal bar-swipe = tab switch · card swipe-to-close · long-press everywhere · never fight Android's predictive back. Every gesture must have a visible-control equivalent. (*Andromeda:  bar-swipe + swipe-up already shipped.*)

Flows 10. Onboarding ≤3 skippable screens: value pitch → import from old browser → default-browser ask. Defer sign-in and permissions to contextual moments. 11. Permission prompts: non-modal chip anchored to the address bar (Chrome found ~85% of prompts get ignored — quiet UI won). 12. Error pages: friendly illustration + plain cause + retry + one moment of delight (Chrome's dino = 270M players/month). (*Andromeda:  "Lost in space" page;  no delight/game yet.*)

4. Cross-browser highlights worth stealing (per browser)

- **Chrome:** tab auto-archiving after N days; tab-group sync; omnibox action chips; Listen-to-this-page; quieter permission chip; incognito biometric lock (*we have this!*).
- **Firefox:** Collections (save tab-sets as named projects); on-device translation; AI Controls hub (every AI feature behind one master switch — trust-building); top/bottom bar choice in onboarding.
- **Edge:** Read Aloud + Immersive Reader + Translate as one reading stack; customizable menu; lesson in what NOT to do (feature churn — Collections/Drop/Sidebar all killed; bloat reputation).
- **Samsung Internet:** drag-and-drop toolbar + menu customization (7 slots); Secret mode with its own anti-tracking level; Video Assistant; Quick Exit panic button; settings search.
- **Brave:** Shields panel anatomy (per-site toggle + counters + advanced expander); aggressive/standard block levels; Forgetful Browsing (auto-logout per site); element zapper; NTP privacy stats as brand reinforcement.
- **Opera:** three tab-switcher layouts; Tab Islands auto-grouping; 100-tab undo; cookie-banner auto-dismitter; Flow (QR-paired, accountless device-to-device sharing).
- **Vivaldi:** Notes in the browser; full-page screenshot capture; per-site blocking levels; "AI-free" as an actual marketed feature.
- **DuckDuckGo:** Fire Button (one-tap wipe with fireproofed exceptions) — the most memorable privacy ritual in any browser; proof a WebView browser can be a serious product.

- **Comet/AI wave:** bottom assistant drawer over the page (not a separate screen); long-press link → "Ask AI"; colored screen border + step log during agent runs; Library of past AI conversations.

5. The WebView boundary — honest engineering limits

From the desktop-superset analysis ([browsers/10-desktop-superset.md](#)):

❌ **Impossible without switching to a Chromium fork (do not promise):** - Real WebExtensions support (Kiwi/Firefox model) — nearest substitute: user-script injection, positioned as "scripting," never "extensions" - uBlock-Origin-parity blocking (response-body filtering, full webRequest) - Engine-level fingerprint resistance (Brave farbling, Firefox RFP) — don't market security we can't implement - DNS-over-HTTPS provider choice, ECH, network-stack controls - Full in-browser DevTools (remote chrome://inspect works; injected Eruda console possible)

✅ **Fully buildable on WebView, and under-served industry-wide (the open lanes):** 1. **Multi-profile / containers** — [androidx.webkit](#) ProfileStore (we already use it for incognito!); almost no mobile browser has real profiles. Firefox containers exist nowhere on mobile. 2. **Desktop-grade tab power** — groups, pinning, tab search, auto-archive, workspaces: all chrome-layer UI. 3. **Own download engine** — parallel segmented, pause/resume, scheduling; beats Chrome Android easily. 4. **Reading list + offline reader** — Readability-extraction + stored clean HTML + Android TTS read-aloud. Pocket's death left a vacuum. 5. **Per-site settings memory** — zoom, desktop mode, dark mode, JS per-origin. Chrome barely does this. 6. **Full-page screenshot + annotation**. 7. **In-app PDF viewer** (PdfRenderer) — most Android browsers punt to external apps. 8. **Brave-lite blocking upgrades** — scriptlets, per-site levels, element zapper, custom filter lists (80% of Brave, honestly labeled). 9. **Omnibox power** — custom search keywords, @tabs/@history scoping, inline calculator, quick actions. No mobile browser does this well. 10. **Assistant-tier AI** — summarize/ask-the-page via cloud API or on-device Gemini Nano (ML Kit GenAI). \$0.001–0.01/page with efficient models; hybrid on-device+cloud is the 2026 pattern. 11. **Translate** — ML Kit on-device translation + DOM rewriting (imperfect on dynamic pages, works). 12. **Media polish** — PiP for fullscreen video, MediaSession notification, background audio.

6. Andromeda v2.0 gap matrix — "the 5%" measured

Legend: ✅ have · 🟡 partial/basic · ❌ missing (buildable) · 🚫 engine-impossible on WebView

6.1 Navigation & address bar

Feature	Status	Notes
Combined omnibox + suggestions (history/bookmarks/web)	✅	v2 P3
Bottom-anchored bar	✅	Orbit Command Bar — matches 2026 consensus
Top/bottom position choice	❌	consensus says offer the toggle
Voice search	❌	RecognizerIntent — easy
QR scanner in bar	❌	deferred from v2; ML Kit
Omnibox quick actions / @-scoping / calculator	❌	open lane, no one does it well on mobile
Copy-URL / share from bar long-press	🟡	share exists in menu

6.2 Tabs

Feature	Status	Notes
Grid thumbnail switcher	✅	v2 P2
Bar-swipe tab switch + swipe-up switcher	✅	v2 P2
Incognito pane separation + theme shift	✅	negative-id design
Tab groups (named/colored)	❌	now table stakes (Chrome/Edge/Samsung/Firefox/Opera all have it)
Tab search	❌	easy win
Recently-closed / undo close	❌	Opera keeps 100
Inactive-tab auto-archive	❌	Chrome's declutter
Pinned tabs / tab lock	❌	
List-view option	❌	consensus: grid default + list option

Feature	Status	Notes
Multi-select bulk actions	✗	

6.3 Home / new-tab

Feature	Status	Notes
Clean search-first home, no feed	✓	matches anti-feed consensus
Bookmarks speed dial	●	v1; not freely editable/orderable shortcuts
Editable shortcut grid (pin/reorder/remove)	✗	the accepted NTP pattern
Wallpapers / home customization	✗	
Privacy-stats block (Brave-style)	✗	we already count blocked ads — surfacing = free brand value

6.4 Privacy & security

Feature	Status	Notes
Incognito + biometric lock	✓	v2 P5 — Chrome-parity
ProfileStore cookie isolation	✓	feature-gated
HTTPS-only mode	✓	v2 P5
Site permission prompts	✓	v2 P5
JS/cookie toggles, clear data, SSL warning	✓	v1
Safe Browsing	✗	setSafeBrowsingEnabled — nearly free win
Tracker blocking as distinct tier + per-site dashboard	●	ad block exists; no tracker-specific lists/panel
Cookie-banner auto-dismiss	✗	Opera/Firefox/Brave have it; filter-list based
Forgetful browsing (per-site auto-forget)	✗	Brave's; CookieManager + storage clear
Fire-button style panic wipe / Quick Exit	✗	DDG/Samsung ritual feature
Global Privacy Control header	✗	header injection — trivial
Fingerprint resistance	⊖	engine-bound — do not market
DoH provider choice	⊖	system Private DNS only
Multi-profile / containers	✗	ProfileStore makes us uniquely positioned — top differentiator candidate

6.5 Ad/content blocking

Feature	Status	Notes
Domain filter lists (EasyList) + exceptions	✓	v1 P4
Generic cosmetic ## rules	✓	v2 P4
Per-site allowlist + counter	✓	
Path/regex rules, \$options	✗	ad-block v2 backlog
Per-site blocking levels (standard/aggressive/off)	✗	Brave/Vivaldi pattern
Element zapper (tap-to-hide)	✗	Brave was first on mobile — high wow
Custom filter-list subscription + user rules	✗	
Scriptlet injection (anti-adblock defense)	✗	partial feasibility
Filter-list auto-update	✗	ours is a bundled snapshot
uBlock parity / response filtering	⊖	engine-bound

6.6 Passwords & autofill — the biggest daily-driver hole

Feature	Status	Notes
Password save/fill	✗	Android Autofill framework integration is the pragmatic path
Passkeys	✗	Credential Manager API

Feature	Status	Notes
Address/payment autofill	✗	
Breach alerts	✗	HIBP-style API

6.7 Downloads & media

Feature	Status	Notes
System DownloadManager + downloads screen	✓	v1 P5
Long-press image/link download	✓	
Own engine: pause/resume/parallel/schedule	✗	deferred twice; fully feasible
In-app file previews	✗	
Background video/audio playback	✗	Brave's biggest Android draw
Picture-in-Picture	✗	needs onShowCustomView handling
Video assistant (floating controls, gestures)	✗	Samsung signature
Offline pages (save full page)	✗	saveWebArchive

6.8 Reading & content

Feature	Status	Notes
Reader mode	✓	v2 P4 (minor leakage known)
Find in page	✓	v2 P3
Desktop-site toggle	✓	v2 P3
Reading list (save-for-later + offline)	✗	Pocket vacuum; pairs with our reader
Full-page translate	✗	ML Kit on-device
TTS / read-aloud	✗	Android TTS + our reader extraction
PDF viewer in-app	✗	industry-wide mobile gap
Print / save-as-PDF	✗	WebView print adapter — easy
Per-site zoom/text-size memory	✗	

6.9 Customization

Feature	Status	Notes
Light/dark/system theme	✓	
Force-dark websites	✓	v2 P4
Own design language (Orbit)	✓	brand asset
Toolbar customization (even 1 slot)	✗	consensus compromise
Menu customization	✗	Samsung/Brave/Edge pattern
Text scaling / force-zoom	✗	accessibility table stakes
High contrast	✗	

6.10 Sync & account

Feature	Status	Notes
Any sync	✗	needs our own backend — v3+ decision; accountless QR-pair (Brave/DDG model) is the trust-friendly route
Bookmark HTML import/export	✓	v2 P6 — the manual fallback
Backup/restore to file	✗	deferred from v2; no backend needed
Send-to-device / Flow-style sharing	✗	backend-dependent

6.11 AI

Feature	Status	Notes
Summarize page	✗	assistant tier — feasible on-device (Gemini Nano flagships) or cloud
Ask-the-page	✗	
AI omnibox answers	✗	
Agentic browsing	✗	industry-wide fragile + prompt-injection unsolved — not recommended
AI master off-switch (Firefox AI Controls)	n/a	if we ever add AI, ship the off-switch with it

6.12 Extras

Feature	Status	Notes
Friendly error page	✓	"Lost in space"
Offline-game / delight moment	✗	dino lesson: cheapest brand surface
Onboarding flow	✗	we have none; ≤3 screens + import + default-ask
Import from other browser at first run	✗	only manual HTML import
PWA / add-to-home-screen	✗	shortcut-level feasible
Full-page screenshot + annotate	✗	Vivaldi/Edge pattern
Settings search	✗	once settings grow
App shortcuts (launcher long-press)	✗	trivial manifest addition

6.13 Scorecard

Category	Coverage vs 2026 daily-driver bar
Navigation & bar	● ~55%
Tabs	● ~40% (structure ✓, management depth ✗)
Home	● ~40%
Privacy & security	● ~55% (strong for our size; missing Safe Browsing, banner-blocker, profiles)
Ad blocking	● ~45% (domain+cosmetic done; levels/zapper/custom missing)
Passwords & autofill	✗ 0% — single biggest gap
Downloads & media	● ~30%
Reading & content	● ~35%
Customization	● ~45%
Sync	✗ ~5% (manual export only)
AI	✗ 0% (deliberate, per owner: v3 = power browser)
Extensions	⊖ engine-bound (user-scripts lane open)
Accessibility	✗ ~15%
Extras/delight	● ~30%

Overall: ~25–30% of the 2026 daily-driver feature bar — with the right skeleton. The v2 architecture (bottom bar, WebViewHolder, ProfileStore, filter engine, Room/DataStore) is exactly the right foundation for the missing 70%; almost nothing needs re-architecting.

7. Strategic readout for v3 (input to discovery — decisions belong to the owner)

Ranked by (user-felt impact × feasibility on WebView × differentiation):

Tier 1 — daily-driver credibility (the features users touch every session): 1. Tab management depth: groups, search, recently-closed undo, list view, bulk actions, auto-archive 2. Passwords & autofill (Android Autofill + passkeys via Credential Manager) 3. Own download engine (parallel, pause/resume, previews) — deferred twice, now due 4. Text scaling / force-zoom + per-site settings memory 5. Safe Browsing toggle + cookie-banner auto-dismiss + editable home shortcut grid

Tier 2 — the "never leave" layer: 6. Reading stack: reading list + offline + translate (ML Kit) + read-aloud (TTS) on top of our reader 7. Ad-block v2: per-site levels, element zapper, custom lists, auto-updating lists, path rules 8. Media: background playback + PiP (Brave's single biggest Android draw) 9. PDF viewer + print/save-as-PDF 10. Backup/restore to file (no backend), onboarding + import flow, QR scanner, voice search

Tier 3 — differentiators nobody on mobile does well: 11. **Profiles/containers via ProfileStore** — genuinely rare; we already ship the API for incognito 12. Omnibox power features (@tabs/@history, custom search keywords, calculator) 13. Toolbar/menu customization (start with one configurable slot) 14. Full-page screenshot + annotate; Quick Exit/panic wipe; privacy stats on home 15. Sync (accountless QR-chain model) — biggest lift, defer until the above land

Explicitly not for v3: extensions (🚫 engine), fingerprint-resistance claims (🚫), agentic AI (industry-proven fragile + insecure), content feeds (industry-proven resented), crypto/rewards (bloat lesson from Brave/Opera reviews).

8. Sources & method

Ten parallel research agents (July 12, 2026), each web-searching vendor release notes, support docs, UX teardowns, and 2024–2026 tech press. Full per-browser detail with source links: [browsers/01-chrome.md](#) · [browsers/02-firefox.md](#) · [browsers/03-edge.md](#) · [browsers/04-samsung-internet.md](#) · [browsers/05-brave.md](#) · [browsers/06-opera.md](#) · [browsers/07-vivaldi-duckduckgo-kiwi.md](#) · [browsers/08-ai-browsers.md](#) · [browsers/09-uiux-patterns.md](#) · [browsers/10-desktop-superset.md](#)

Chrome (Android)

1. Overview & positioning

Chrome is the default, preinstalled browser on virtually all Google-certified Android devices and the dominant mobile browser worldwide: roughly 65% of global mobile browser share as of spring 2026 (Statcounter), with the Chrome-for-Android variant alone accounting for ~60% of mobile sessions; Samsung Internet (~2.8%) is its nearest Android rival. It runs on the Blink engine (Chromium) with Google's V8 JS engine and targets the mainstream user deeply invested in the Google ecosystem — sign-in, Password Manager, autofill, Search, Discover, and now Gemini are all first-class. Positioning in 2025–2026 has shifted hard toward "AI-first agentic browser": Gemini in Chrome, auto browse, and AI Mode search are the headline bets, layered on an otherwise conservative, fast, minimal browser.

2. Feature inventory by category

Navigation & address bar (omnibox)

- Combined search/URL omnibox with Google Search suggestions, trending searches, and inline answers (weather, sports, conversions) before you finish typing
- **Configurable position:** address bar can be moved to the bottom of the screen (long-press the bar → "Move address bar to bottom", or Settings → Address bar) — rolled out mid-2025
- Mic (voice search, Assistant-style UI) and Google Lens camera icons inside the bar/new-tab search box
- **AI Mode chip** in the omnibox that hands your query to Google's conversational AI Mode search
- Gemini logo/entry in the address bar area for invoking Gemini in Chrome and connected Workspace apps (Docs, Drive, Keep, Calendar) — 2026, US
- Chrome Actions: typed commands surface action chips ("clear browsing data", "open incognito tab") plus local-business chips (call, directions, reviews)
- Contextual suggested questions about the current page; "related searches" chips
- Swipe left/right on the address bar to switch tabs; swipe down (older gesture) for switcher
- Copy-link, share, and site-info accessible from the bar; security/tune icon shows permissions
- Touch to Search: tap a word on a page to get a slide-up Google search panel for it
- Search-ready omnibox: focusing the bar offers your recent/related searches and clipboard URL suggestion

Tabs & tab management

- Grid-based tab switcher with card thumbnails; incognito/regular panes; "+" new tab; : menu (close all, select tabs)
- **Tab groups:** drag one tab card onto another to group; named + colored groups; groups render as stacked cards and as a bottom tab-strip row when a grouped tab is open (controls tweaked in Chrome 138)
- **Tab group sync** across devices and group restore (Feb 2025)
- **Tab search** within the switcher (Feb 2025)
- **Inactive-tab archiving ("tab declutter"):** tabs unused for 21 days (configurable 7/14/21, or off) auto-move to an Archived section; duplicate tabs auto-archived; optional auto-delete of archived tabs after 60 days; manual drag-to-archive in development
- **Pinned tabs** on Android (late 2025), shrink to the left of the strip/top of grid; tablet drag-out keeps pin
- Tab strip on tablets/foldables with revamped active-tab highlighting; drag tabs between windows; multi-window/multi-instance support
- Recent tabs menu shows tabs open on your other signed-in devices
- Reopen recently closed tabs; long-press the tab-switcher button for quick New tab/Close tab menu
- Bulk-select tabs to group, share, or close

Home / New-tab page

- Google logo + centered search box (with mic + Lens), grid of most-visited/shortcut tiles
- **Discover feed:** personalized news/content feed below the fold; can be turned off ("Customize new tab page")
- Cards: resume "Continue browsing" tiles, price-drop cards (US), and other segments — individually toggleable
- Customization menu: show/hide shortcuts, toggle feed/cards, custom background image or theme color
- Optional separate Homepage (any URL) distinct from the NTP, set in Settings

Search & discovery

- Google default but changeable engine (recently visited engines are offered); EU users get a DMA choice screen
- **Google Lens**: search what's on screen/camera, image long-press "Search image with Google", visual shopping matches
- Circle to Search integration on Android (long-press home/nav bar works over Chrome content)
- Voice search; AI Mode conversational search; trending searches; shopping-category image previews in suggestions
- Translate-integrated search; Touch to Search word lookup
- Journeys-style resumable search/browsing history clusters feed "resume" suggestions

Privacy & security

- Safe Browsing: Standard + **Enhanced Protection** (real-time URL checks, deep scans); auto-running **Safety Check** (compromised/weak/reused passwords, permission review, Safe Browsing reminders, notification-spam review)
- **Incognito mode** with third-party cookies blocked by default; **Incognito lock** (biometric re-auth to resume sessions); Incognito screenshot blocking option
- **IP Protection** in Incognito (proxying to mask IP from known trackers, from Q3 2025)
- Third-party cookies retained in regular browsing (Google abandoned deprecation); Privacy Sandbox ad controls (Ad topics, Site-suggested ads, Ad measurement) under "Ad privacy"
- Privacy Guide walkthrough; Delete browsing data quick action (15-min quick delete)
- HTTPS-First options; Secure DNS (DoH); Always use secure connections toggle
- One-time (this-time-only) camera/mic permission grants; auto-revocation of permissions/notifications from unused or abusive sites
- Google Password Manager: on-device encryption option, breach alerts, biometric fill, **passkey** creation/storage synced via GPM
- Site settings: per-site permissions (camera, mic, location, notifications, pop-ups, JS, sound, storage, NFC, motion sensors, etc.); new consolidated **Site controls** menu entry (2026)
- Prompt-injection defenses and confirmation gates around agentic (auto browse) actions

Ad/content blocking

- No extension support, so no real ad blockers — the biggest gap vs. Firefox/Edge/Samsung Internet
- Built-in heavy-ad and intrusive-ad filtering per Better Ads Standards (blocks ads sitewide on violating sites)
- Pop-up and redirect blocking on by default; abusive-notification prompt blocking and one-tap unsubscribe from site notifications
- Quieter permission prompts; that's the ceiling — users wanting uBlock-class blocking must use Private DNS/VPN blockers or another browser

Downloads & media

- Built-in download manager (Downloads home) with in-progress notification, pause/resume; parallel downloading available via flag
- "Download page" for offline reading; offline badge suggestions; auto-resume downloads when back online
- **Native inline PDF viewer** (2025): open PDFs in-browser with pen/highlighter markup, eraser, undo/redo, "Save copy", find, and a **Save to Drive** button (2026); no signing yet
- Video: picture-in-picture support, media-session notification controls, background playback for PiP; media casting to Chromecast devices via Cast
- Long-press link → "Download link"; image → "Download image"
- Smart handling of file types (opens PDFs inline instead of forcing download)

Reading & content

- **Reading mode**: distraction-free article view (was a bottom sheet, redesigned to a full-page experience in late 2025/2026) with font, size, spacing, and color controls
- **Listen to this page**: TTS with mini-player (speed, voices, highlighting, auto-scroll, background/tab-switch listening); **AI playback** turns articles into podcast-style audio; ~12+ languages
- Google Translate built in: full-page translation bar, partial-selection translate, auto-offer, per-language preferences; combine with Listen for translated audio
- Reading list (save for later, offline-readable) integrated with bookmarks
- Find in page; desktop-site toggle (global + per-site)
- Print / save as PDF from Share menu

Customization & appearance

- Theme: system/light/dark; websites can be asked for dark theme (auto-darken flag)
- NTP customization (background image, shortcuts, feed toggles)
- Address bar top/bottom placement
- **Toolbar shortcut**: configurable action button in the top toolbar (new tab / share / voice search or auto-picked)
- Material 3 Expressive redesign (Chrome 141, Oct 2025): rounded buttons, pill toolbars, refreshed menus
- Homepage URL; search engine choice; per-site zoom defaults
- No full theming/skin engine and no bottom-menu option — customization remains shallow vs. rivals

Sync & account

- Modern **sign-in without "sync"** model (2024+): one-tap Google sign-in gets passwords, payment methods, addresses, bookmarks, reading list, settings; **history + open tabs sync is a separate opt-in**
- Sync across desktop/iOS/Android: tabs, groups, history, passwords/passkeys (GPM), autofill, settings
- Send to your devices (push a tab to another signed-in device)
- Account-level controls: Web & App Activity ties into personalization; per-datatype sync toggles; encryption passphrase option

AI features

- **Gemini in Chrome for Android** (June 2026, US, Android 12+, 4GB+ RAM): summarize pages, ask questions about page content, cross-app actions with Calendar/Keep/Gmail without leaving Chrome
- **Auto browse** (AI Pro/Ultra subscribers): agentic task automation — book parking, reorder items, fill forms — with confirmation prompts for sensitive actions and prompt-injection safeguards; ties into Gemini Spark for 24/7 background agent actions
- **Nano Banana** image generation/customization in-browser (e.g., turn a page into an infographic, edit images found on the web)
- **AI Mode** chip in the omnibox; contextual suggested questions about the current page
- AI playback (podcast-style Listen); AI-organized history/search journeys
- On-device Gemini Nano powers features like Scam-detection warnings in Safe Browsing
- Note: some Gemini-in-Chrome extras (Skills/saved prompts, cursor element selection) launched desktop-first

Gestures & shortcuts

- Swipe left/right on the address bar to cycle tabs
- Pull-to-refresh (not user-disableable, a perennial complaint)
- Android system back gesture with predictive back; in-page back/forward history navigation via edge swipes (overscroll history navigation)
- Long-press back button → full history stack for that tab
- Long-press tab-switcher icon → New tab / Close tab menu; swipe down from toolbar into switcher
- Drag-down through the : menu in one motion to select an item
- Long-press address bar → move-to-bottom option; long-press reload → request desktop site (older builds)
- Drag tab cards onto each other to form groups; drag to reorder; (in dev) drag to archive
- App shortcuts from launcher icon long-press: New tab, Incognito tab, Search, Dino game

Data saving & performance

- Lite Mode/Data Saver was **discontinued (2021)** — no compression proxy remains
- Preload pages setting: No / Standard / Extended preloading (Extended routes predictions through Google)
- Back/forward cache for instant back navigation; freezing/discarding of background tabs to save memory
- GPU rasterization, QUIC/HTTP3 by default; flags for parallel downloading and rendering tweaks
- Auto-archiving of inactive tabs reduces memory bloat
- No user-facing "memory saver"/"battery saver" toggles like desktop; still criticized as RAM/battery heavy
- chrome://flags remains the power-user tuning surface (note: 2026 advice includes disabling the ~4GB on-device Gemini model download via #optimization-guide-on-device-model on storage-constrained devices)

Extensions/add-ons

- **No extension support on phones/tablets** — Chrome's most cited Android weakness

- Google is building **desktop-class Chrome builds for Android-powered PCs** that install and persist Chrome Web Store extensions (experimental, 2025–2026); no announced plan to bring this to phones
- Users wanting extensions use Edge, Firefox, Quetta, Yandex, or (formerly) Kiwi

Accessibility

- Page zoom control (per-site and default zoom levels, up to 300%+), force-enable zoom on sites that block it
- Simplified view for accessibility / reader mode; text scaling
- TalkBack screen-reader support, Switch Access, Voice Access compatibility
- Listen to this page (TTS) doubles as an accessibility feature; OS-level Live Caption works over Chrome media
- Image descriptions from Google (get descriptions of unlabeled images, on-demand or always for screen-reader users)
- High-contrast/dark theme, forced dark mode for web contents (flag)

Unique extras

- **Chrome Custom Tabs** platform feature: in-app browsing surface for other Android apps, incl. **minimized custom tabs** (shrink an in-app page to a floating PiP chip) and partial-height/side-by-side custom tabs
- PWA support: installable web apps via WebAPK ("Install and create shortcut"), Add to Home screen, Web Push, TWA
- QR code generator/scanner in the Share sheet; built-in long-screenshot capture and editor (crop, text, draw) on some builds
- Cast to Chromecast; Print; Send to your devices; Copy link — all in Chrome's own share sheet
- Price tracking / price-drop cards for tracked products (US)
- Dino offline game (chrome://dino)
- chrome:// internals pages (flags, version, net-internals) all work on Android
- First-party Google integration: Discover feed, Lens, Translate, Password Manager, Family Link supervision, enterprise policy management

3. UI/UX structure & feature placement

- **Address bar position:** top by default, **user-movable to bottom** (long-press bar or Settings → Address bar). Inside it: site security/tune icon (page info/permissions), URL/search field, mic, Lens (on NTP box), AI Mode chip contextually, then toolbar-level: optional configurable shortcut button, tab-switcher square with tab count, : menu. Gemini entry appears in/near the bar on eligible 2026 US devices.
- **Tab switcher:** grid of rounded card thumbnails with favicon+title and close X; reached by tapping the tab-count square (or address-bar swipe-down gesture). Tab groups show as stacked/merged cards that open into a dedicated group view with name and color; toggle between regular and Incognito panes; toolbar has "+" (new tab), search-tabs icon, and : (Close all tabs, Select tabs); an "Archived tabs" entry surfaces auto-archived inactive tabs. Tablets/foldables get a persistent top tab strip instead.
- **Main : menu (post-Chrome 150 redesign, mid-2026),** roughly in order: icon row at top — Back (new), Forward, Bookmark/star, Download, Refresh (page-info icon removed from row); then list: New tab · New Incognito tab · (New window, tablets) · Ask Gemini (eligible devices) · History · Delete browsing data · Downloads · Bookmarks · Recent tabs · Share · Find in page · Listen to this page · Translate · Install and create shortcut (renamed from "Add to Home screen") · Desktop site toggle · Site controls (new consolidated per-site permissions entry) · Settings · Help & feedback. (Exact composition varies by server-side rollout; pre-150 builds show Page info in the icon row and "Add to Home screen".)
- **Settings screen (post-April 2025 redesign),** top-level in order: **You and Google** (sign-in card, sync/account services) → **Basics:** Search engine · Address bar · Privacy and security · Safety check → **Passwords and autofill:** Google Password Manager · Payment methods · Addresses and more · Autofill services → **Advanced:** Tabs · Homepage · New tab page cards · Toolbar shortcut · Notifications · Theme · Accessibility · Site settings · Languages · Downloads · About Chrome.
- **Long-press / context menus:** on a link — rich sheet with favicon/title header, Open in new tab · Open in new tab in group · Open in Incognito tab · Preview page (overlay peek) · Copy link address/text · Download link · Share link; on an image — thumbnail header plus Open image in new tab · Download image · **Search image with Google (Lens)** · Copy/Share image (link options separated by a divider if the image is also a link); on selected text — Copy, Share, Select all, Translate, Web search; on the address bar — copy/paste options plus "Move address bar to bottom/top". Home-screen icon long-press gives New tab / Incognito tab / Search / Dino shortcuts.
- **Gestures:** pull-to-refresh; horizontal swipe on address bar to change tabs; swipe down from toolbar into tab switcher; overscroll/edge swipe back-forward history navigation with predictive back; long-press back for tab history; drag-to-group tabs in switcher; drag menu open-and-select in one motion.
- **First-run/onboarding:** splash → single modern Google-account sign-in sheet (post-2024 "sign in, no sync toggle" flow; separate opt-in later for history/tab sync) with "Use without an account" escape → optional Enhanced ad privacy (Privacy Sandbox) consent notice → notification permission prompt (Android 13+) → lands on NTP with a "touch and hold to move the address bar" education prompt where

applicable. EU/DMA devices insert default-browser and search-engine choice screens. Default-browser prompting otherwise relies on the Android system dialog.

4. Standout features (what Chrome does best)

- Deepest Google integration anywhere: Search/Lens/Translate/Discover/Password Manager/passkeys/Workspace in one surface
- The 2026 AI stack: Gemini in Chrome, auto browse agentic tasks, Nano Banana image generation, AI Mode — far ahead of other Android browsers' AI depth
- Best-in-class sync (now friction-free sign-in model) across the world's largest browser install base
- Tab lifecycle management: groups + group sync + search + auto-archiving/declutter is arguably the most complete mobile tab hygiene system
- Listen to this page with AI podcast-style playback and translated audio
- Performance and web-standards leadership (Blink/V8, QUIC, bfcache) — consistently fast page loads and top compatibility since sites are built for Chrome
- Safety stack: Enhanced Safe Browsing, auto Safety Check, Incognito lock, IP Protection, on-device scam detection
- Custom Tabs/WebAPK platform plumbing that makes it the substrate of Android in-app browsing

5. Weaknesses & common criticisms

- **No extensions on phones** — no uBlock-class ad blocking, no user scripts; desktop-Android extension work explicitly excludes phones for now
- Privacy reputation: data flows to Google (Extended preloading, signed-in defaults, Privacy Sandbox ad targeting, Gemini page-content processing); Incognito misconceptions lawsuit legacy
- RAM and battery hunger, especially with many tabs on budget devices
- Ad blocking limited to "intrusive ads" standards — Chrome is structurally conflicted as Google's ad-delivery surface
- Server-side/staged rollouts mean features (menus, bottom bar, Gemini) appear inconsistently across users and regions; many 2026 AI features are US/English-only and some are paywalled (auto browse behind AI Pro/Ultra)
- Customization shallow vs. Samsung Internet/Firefox/Vivaldi: no bottom menu, no full theming, pull-to-refresh not toggleable
- Late to features rivals had for years (bottom address bar, PDF viewer, pinned tabs)
- Creeping AI clutter (AI Mode chip, Gemini entries) and silent multi-GB on-device model downloads criticized by power users
- Antitrust/DMA scrutiny of default status; monoculture concerns about Blink dominance

6. Recent additions (2024–2026)

- **2024:** sign-in-without-sync account model; Listen to this page TTS (Chrome 125); tab declutter/auto-archive of inactive tabs; minimized Custom Tabs (floating PiP web windows); Chrome Actions on mobile + local-business action chips; trending-search and shortcut suggestions in omnibox; shopping-category image suggestions; real-time Enhanced Safe Browsing; tab group improvements and cross-platform group work; Circle to Search interplay
- **2025:** tab search and tab group sync (Feb); native inline PDF viewer with markup (Apr); Settings reorganized into You and Google / Basics / Passwords and autofill / Advanced (Apr); bottom address bar rollout (Jun–Jul); tab-group control tweaks (Chrome 138); IP Protection in Incognito (Q3); Material 3 Expressive redesign (Chrome 141, Oct); pinned tabs on Android; tablet/foldable tab-strip revamp; Reading Mode redesign (Dec); Safety Check auto-run expansion
- **2026:** full-page Reading Mode; Save to Drive button in PDF viewer; **Gemini in Chrome for Android** (summaries, Q&A, Workspace hooks — late June, US); **auto browse** agentic automation (AI Pro/Ultra); **Nano Banana** in-browser image generation; AI Mode omnibox chip broadened; Chrome 150 menu redesign — dedicated Back button in icon row, new "Site controls" entry, "Install and create shortcut" rename (Jul); WebMCP origin trial for browser agents (Chrome 149); experimental desktop-class Chrome for Android PCs with Chrome Web Store extension installs

Firefox (Android)

1. Overview & positioning

Firefox for Android (codebase "Fenix") is the only major Android browser built on a non-Blink engine — Mozilla's **GeckoView** — making it the sole mainstream alternative rendering engine on the platform and the only major Android browser with an **open extension ecosystem**. Market share is small: roughly **0.6% of worldwide mobile sessions** (Statcounter, April 2026) and ~1.4% of US mobile, versus ~2.2–2.9% cross-platform — Firefox's base skews heavily desktop. Target user: privacy-conscious and power users who want uBlock Origin-class ad blocking, deep tracking protection, cross-device sync outside Google's ecosystem, and customization (address bar position, wallpapers, extensions). Positioning is "independent, private, yours" — explicitly anti-surveillance-capitalism, with 2026 messaging emphasizing *user-controlled, opt-in AI*.

2. Feature inventory by category

Navigation & address bar

- Combined URL/search bar with search suggestions, history/bookmark/open-tab autocomplete
- Address bar placeable at **top or bottom** (bottom is the modern default for new installs)
- Auto-hide toolbar on scroll (toggleable in Settings > Customize)
- QR-code scanner built into the address bar
- Voice search microphone in the address bar
- Search-engine shortcuts (tap to search with a one-off engine from the address bar)
- Optional dedicated navigation bar (back/forward/home/tabs) added with the 2025 toolbar redesign; back/forward arrows on toolbar with long-press for tab history
- "Refreshed toolbar" (v148, Feb 2026): cleaner look plus toolbar customization options
- Shield icon in address bar shows Enhanced Tracking Protection status per site; padlock for permissions/connection info
- Summarize-page entry point in the address bar (152, where eligible)

Tabs & tab management (incl. Collections)

- Tab tray (opened via numbered tab counter) with **List or Grid view**
- Three tab-tray pages: **normal tabs**, **private tabs**, **synced tabs** (from other devices)
- **Tab Groups** — color-coded, nameable groups; rolled out to Android with v152 (June 2026) after a Mozilla Connect co-design program; groups can be collapsed/reopened
- **Collections** — long-standing Fenix feature: save sets of open tabs into named collections (e.g., Travel, Shopping) that live on the home screen; add/rename/share/reopen; distinct from tab groups (collections are saved snapshots, groups are live tabs)
- **Inactive tabs** — tabs untouched for ~14 days auto-move to a collapsible "Inactive" section with one-tap close-all
- Close tabs automatically after one day/week/month (Settings > Tabs)
- Select multiple tabs (long-press) to move to collection, share, or close
- "Recently closed tabs" list under History; undo close-tab snackbar
- Swipe a tab in the tray to close it; "Close all tabs" in tray menu
- Private tabs wiped on exit option; private-browsing shortcut on Android home screen

Home / New-tab page

- Firefox Home shows: **Shortcuts** (frequent sites, pinnable), **Sponsored shortcuts** (ads, can be disabled), **Recently visited**, **Recent bookmarks**, **Jump back in / recent tabs**, **Collections**, and **Stories/recommended articles** (Pocket-derived recommendations, English locales)
- Every section individually toggleable via Settings > Homepage / "Customize homepage"
- **Wallpapers** for the home screen (Classic + Limited Edition sets; no custom image from gallery)
- Opening-screen behavior setting: always Home, last tab, or Home after four hours of inactivity
- "You're protected" tracker-blocking message on home (tweaked in 152.0.4 to appear only after trackers actually blocked)
- World Cup live-scores widget on Home (151.0.4/152, June 2026)

Search & discovery

- Multiple pre-installed engines (Google default revenue deal; DuckDuckGo, Bing, etc.); **add custom search engine** by URL template
- Separate default engine choosable for private browsing; toggle to show/hide search suggestions in private mode
- Search widget for the Android launcher home screen
- "Search" via text-selection context menu
- Autocomplete from history/bookmarks/clipboard ("fill link from clipboard")
- Firefox Suggest-style sponsored suggestions largely a desktop feature; Android keeps suggestions simpler

Privacy & security (ETP, etc.)

- **Enhanced Tracking Protection** with Standard / Strict / Custom modes, per-site shield toggle, and Total Cookie Protection (cookie isolation per site)
- **Private browsing** with dedicated tab tray page, mask theming, optional lock (device auth), and no summaries/telemetry
- **HTTPS-Only Mode** (all tabs or private-only)
- **DNS over HTTPS** user-facing setting (surfaced v143, Sept 2025): Default/Increased/Max protection with provider choice
- **Cookie Banner Blocker** ("Reduce cookie banners") auto-rejects consent popups
- **Global Privacy Control** toggle
- **Site Isolation (Fission)** on Android from v147 (Jan 2026) — Spectre-class side-channel protection matching desktop
- **Certificate Transparency enforcement** and **CRLite** local revocation checking (v145, Nov 2025)
- **Post-quantum crypto**: ML-KEM (mlkem768x25519) in TLS 1.3/HTTP3 (v145)
- Delete browsing data on quit; granular "Delete browsing data" (tabs, history, cookies, cache, permissions, downloads)
- Password manager with breach-alert integration (Firefox Monitor lineage), biometric gate, autofill; third-party autofill service support
- Autofill for addresses and payment methods; sync of logins/addresses/cards (E2E encrypted)
- Tracker-blocking counts per site via the shield panel; SmartBlock unbreaks sites in Strict mode
- Remote-improvements opt-in decoupled from telemetry (v148) — can receive remote config without sending data

Ad/content blocking

- **No built-in ad blocker** — by design Mozilla relies on ETP (blocks trackers, cryptominers, fingerprinters, tracking cookies/content in Strict)
- Full **uBlock Origin (Manifest V2)** available as an extension — the flagship reason many users choose Firefox on Android; Chrome Android offers no extensions at all
- Other blockers: AdGuard AdBlocker, Ghostery, Privacy Badger, NoScript, Consent-O-Matic, SponsorBlock for YouTube

Downloads & media (background video/PiP)

- Revamped **download manager** (v143): real-time progress tracking, pause/cancel quick controls, notification integration
- **Background audio/video playback** (continues with screen off via media notification) — notably works for YouTube with extensions like "Video Background Play Fix"
- **Picture-in-Picture** video support (system PiP)
- Autoplay blocking (block audio only / block audio & video)
- Media session controls in notification shade / lockscreen
- Open PDFs inline (PDF.js); **share the PDF file itself or copy a link from the address bar** (v152); save page as PDF / print
- xHE-AAC high-quality audio codec support (v143)
- Downloads accessible from menu quick-row; files land in system Downloads

Reading & content (reader mode, translate, TTS)

- **Reader View** (book icon in address bar on article pages): font size/typface, light/sepia/dark themes
- **Built-in full-page translation** — fully **on-device** (no cloud), auto-offers on foreign-language pages, translates continuously as you browse; settings for always/never translate per language/site; Zstandard-compressed models shrink downloads (v145); improved RTL/bidi handling (v145)
- No built-in text-to-speech / read-aloud (users rely on extensions like Read Aloud or Android's system Reading Mode)
- Find in page; copy link text without opening the page (v148)
- "Open in app" handoff and desktop-site toggle per tab

Customization & appearance (address bar position!)

- **Address bar/toolbar position: top or bottom** — the signature customization Chrome long lacked; plus separate navigation-bar option

- Toolbar customization (which actions appear) via the 2025–26 "composable toolbar" work (v148+)
- Auto-hide toolbar toggle
- Theme: Light / Dark / Follow device / set by battery saver
- Home-screen wallpapers; homepage section toggles; shortcuts on/off
- Gestures settings (pull-to-refresh toggle)
- Custom add-on collection support (Nightly) to sideload arbitrary extension sets
- **Secret settings / debug menu**: tap Firefox logo 5x in Settings > About Firefox — exposes feature flags (e.g., disable the new tab tray/toolbar redesigns)
- **about:config available only on Nightly** (not Release/Beta) — a persistent power-user complaint

Sync & account

- Mozilla account sync: bookmarks, history, open tabs, passwords, addresses, payment methods, settings — **end-to-end encrypted**
- **Send tab** to/from any signed-in device; synced-tabs page in the tab tray
- Sign-in via QR code pairing with desktop
- No multi-profile support on Android (profiles shipped on desktop v144, not mobile)

AI features

- **Page summaries / "Shake to Summarize"** (v151–152, May–June 2026): shake the phone, or tap in the address bar / "Summarize page" in the menu — AI summary of the article; runs on **Mistral Small 3.1 via Mozilla's cloud**; adapts output (recipes → steps, sports → scores); English-only, pages under ~5,000 words that support Reader View; unavailable in Private Mode and behind paywalls
- **AI Controls hub** (v151): a single settings surface to turn every AI feature on/off or configure individually — deliberate "AI on your terms" positioning; turning AI off disables the shake gesture and summaries together
- No sidebar chatbot on Android (desktop-only); no AI-injected search results of its own

Gestures & shortcuts

- Pull-to-refresh
- Swipe address bar horizontally to switch between adjacent tabs
- **Swipe vertically across the toolbar** to open the tab view / move between tabs (v151)
- **Shake to summarize** (v152)
- Long-press back/forward arrows for tab history
- Pinch-zoom (performance overhauled in v152, notably on low-end devices)
- Android app shortcuts (long-press launcher icon: new tab, new private tab)
- Custom Tabs support for links opened from other apps (with Firefox menu inside)

Data saving & performance

- No data-saver/compression mode (unlike Opera/Chrome Lite lineage)
- Rust-based networking stack for QUIC + HTTP/3 (v145) for faster connections
- GeckoView-tuned rendering; benchmark reality: ~10–15% slower than Chrome on Speedometer, though roughly comparable in daily feel; some tests show lower per-tab RAM than Chrome
- Minimum Android version raised to 8.0 and 32-bit x86 dropped (v144, Oct 2025)

Extensions/add-ons (which extensions work on mobile)

- **Open extension ecosystem since Dec 2023 (v121)** — any developer can publish Android-enabled extensions to addons.mozilla.org; launched with 450+, grown past 1,000+ by 2025; still far below desktop's ~40,000
- Marquee working extensions: **uBlock Origin, Dark Reader, Bitwarden, Privacy Badger, NoScript, Violentmonkey/Tampermonkey (userscripts), SponsorBlock, Ghostery, Consent-O-Matic, Read Aloud, Video Background Play Fix, Simple Tab Groups, TWP/Linguist translators, Search by Image, ClearURLs, Decentraleyes/LocalCDN**
- Extensions managed via the main menu's Extensions section (with recommendations) and Settings; per-extension private-mode permission
- MV2 preserved — full-strength blocking APIs that Chrome's MV3 curtailed
- Nightly + custom AMO collection trick allows installing almost any desktop extension (unsupported)

Accessibility

- Font size follows Android system settings or custom scaling ("Automatic font sizing")
- **Zoom on all websites** override (force pinch-zoom even where sites block it)
- TalkBack/screen-reader support; voice input; reader view as a de-cluttering aid
- High-contrast/dark theme support; menu redesign (v141) explicitly targeted accessibility/reachability

Unique extras

- Only Gecko engine on Android; only major Android browser with real extensions
- Collections (unique save-tabs-to-project model)
- On-device translation (no cloud) — privacy-unique among major mobile browsers
- Custom Tabs + "Open links in apps" control
- Add to Home screen / PWA install (shortcut-based; no WebAPK, so weaker than Chrome's PWA integration)
- about:debugging via USB for remote devtools inspection of the phone from desktop Firefox (Settings > Remote debugging via USB)
- Firefox Nightly/Beta channels co-installable for early features
- World Cup widget (2026, opt-in home component)

3. UI/UX structure & feature placement

- **Address bar position and contents:** user choice of top or bottom (Settings > Customize; also asked in onboarding). Contains: shield (ETP) icon, padlock/site info, URL/search field, summarize entry (eligible pages), reader-view icon on articles, refresh, tab counter, three-dot menu. Optional separate navigation bar (back/forward/home/tabs/menu) from the 2025 redesign; auto-hides on scroll unless disabled.
- **Tab switcher/tab tray:** tap the numbered tab counter. Redesigned (2025) tray is a full/half sheet with three tabs/pages: **normal**, **private (mask icon)**, **synced**. Grid or List layout (Settings > Tabs). Contains: new-tab FAB/button, three-dot tray menu (select tabs, share all, close all, tab settings, recently closed), Inactive-tabs collapsible section, tab-group chips (152+), banner to save selected tabs to a Collection. Redesign criticized for lower density (~4 tabs visible vs ~8 before).
- **Main menu (three-dot; redesigned v141, July 2025)**, roughly in order: 1. Mozilla account sign-in/sync banner (top) 2. Navigation row: **Back**, **Forward**, **Refresh**, **Share** 3. Quick-access row: **Bookmarks**, **History**, **Downloads**, **Passwords** 4. **Extensions** (expandable; shows installed + recommended add-ons) 5. Page section: **New tab / New private tab** (contextual), **Switch to desktop site**, **Find in page**, **Translate page**, **Summarize page** (152+) 6. **Save** group: Bookmark this page, Add to shortcuts, Add to Home screen, Save to collection, Save as PDF 7. **Tools** group: Reader View, Print, Open in app, Report broken site 8. **Settings** (pinned at the bottom) Older (pre-141) menu was a long flat list; the redesign grouped it with submenus and cut taps for common actions.
- **Settings screen — top-level sections in order** (approximate current structure):
 - **General:** Search · Tabs · Homepage · Customize (theme, toolbar position, gestures) · Passwords · Autofill (addresses, payment methods) · Accessibility · Language · Translations · Set as default browser
 - **Privacy and security:** Private browsing · HTTPS-Only Mode · Reduce cookie banners · Enhanced Tracking Protection · AI controls (151+) · Site permissions · Delete browsing data · Delete browsing data on quit · Notifications · Data collection
 - **Advanced:** Extensions/Add-ons · Remote debugging via USB
- **About:** Rate on Google Play · About Firefox (5 taps on logo = debug/secret settings)
- **Long-press / context menus:**
 - Link: Open in new tab / Open in private tab / Copy link / **Copy link text** (148+) / Share link / Download link / Bookmark link
 - Image: Open image in new tab / Save image / Copy image location / Share image
 - Address bar: paste / paste & go / copy URL; clipboard-fill suggestion chip
 - Tab (in tray): select mode for multi-tab actions; toolbar back/forward long-press = session history
 - Text selection: copy, share, search, translate (and Private Search)
- **Launcher icon:** New tab / New private tab shortcuts
- **Gestures supported:** pull-to-refresh; horizontal swipe on toolbar to change tabs; vertical swipe across toolbar for tab view (151+); shake-to-summarize (152+); pinch zoom (always-allowed option); scroll-to-hide toolbar; swipe-to-close tabs in tray.
- **First-run/onboarding flow:** card-based sequence — (1) welcome + **set as default browser** (deep-links to Android default-apps setting), (2) **choose address bar position (top/bottom)** on newer builds, (3) **sign in to sync** (QR pairing option), (4) notification permission

request, (5) optional add-search-widget card; ends with "Start browsing." Telemetry/data-collection notice with opt-out; onboarding reruns only after clearing app data.

4. Standout features (what Firefox does best)

- **Real extensions on mobile** — uBlock Origin at full MV2 strength; no other major Android browser has an open extension ecosystem
- **Best-in-class tracking protection defaults:** Total Cookie Protection, ETP Strict, GPC, cookie-banner rejection — and ~2x better fingerprinting resistance than Chrome out of the box in third-party tests
- **On-device translation** — private by architecture, no text leaves the phone
- **Address bar position choice + toolbar customization** — long a differentiator vs Chrome
- **Engine independence** (Gecko) — sync ecosystem and rendering un beholden to Google; MV2 ad-blocking APIs preserved
- **Power-user escape hatches:** Nightly with about:config, secret settings, remote USB debugging, custom add-on collections
- **Opt-in, clearly-controlled AI** (AI Controls hub) — contrast with rivals baking AI in with weak off-switches
- **Collections + tab groups + inactive tabs** — arguably the deepest tab-organization stack on Android now
- Background media playback + PiP handled unusually well (esp. with extensions)

5. Weaknesses & common criticisms

- **Tiny market share** (~0.6% mobile worldwide) → some sites/banks UA-sniff for Chrome, occasional compatibility glitches on content-heavy sites (patched repeatedly, e.g., 146.0.1)
- **Performance gap vs Chromium:** ~10–15% slower Speedometer, slower first paint on some sites; historically janky scrolling on low-end hardware (152's pinch-zoom work is a response)
- **Weaker PWA story:** no WebAPK integration; "Add to Home screen" is closer to a shortcut; fewer web-app APIs than Chrome
- **UI redesign backlash (2025–26):** new tab tray density ("4 tabs where 8 fit before"), forced rollout without opt-in, reversion only via secret/debug settings — heavy criticism on Mozilla Connect and r/firefox, including accessibility complaints
- **No about:config on Release** — power users must run Nightly
- **Extension catalog still limited** (1,000+ vs ~40,000 desktop); some flagship desktop extensions lack Android support or proper mobile UIs
- **No built-in ad blocker or data-saver mode** — relies on extensions; heavier out-of-box than Brave for ad-averse users
- Menu/quick-actions redesign polarizing (screen space, muscle-memory breakage)
- Mozilla-org distrust threads: sponsored shortcuts/ads on Home by default, telemetry defaults, AI feature creep (partly answered by AI Controls and the 148 telemetry decoupling)
- Historic Android churn (Fennec→Fenix regression memories) still colors reviews

6. Recent additions (2024–2026)

- **Dec 2023 / through 2024:** open extension ecosystem matured (hundreds → 1,000+ Android add-ons); translation rolled out broadly (v126–130); Global Privacy Control; ongoing menu-redesign experiments
- **2025:**
 - **v141 (Jul):** redesigned main menu (quick-access row, back/forward in menu, expandable Extensions, Settings at bottom)
 - **2025 toolbar/tab-tray overhaul** ("composable toolbar"): new tab manager tray, toolbar customization, share button surfaced, auto-hide toolbar, optional navigation bar
 - **v143 (Sep):** new download manager with live tracking; user-facing **DNS over HTTPS** settings; xHE-AAC audio
 - **v144 (Oct):** min Android 8.0, 32-bit x86 dropped; tab-group groundwork
 - **v145 (Nov):** Certificate Transparency enforcement, **CRLite**, **ML-KEM post-quantum TLS**, Rust QUIC/HTTP3 stack, lighter Zstandard translation models
- **2026:**
 - **v147 (Jan): Site Isolation (Fission)** on Android — desktop-grade Spectre protection
 - **v148 (Feb):** refreshed toolbar look + customization; **copy link text**; remote improvements decoupled from telemetry
 - **v150 (Apr):** stability/smoothness maintenance release
 - **v151 (May): AI Controls hub**; vertical toolbar-swipe tab gesture; **page summaries** rollout begins
 - **v151.0.4 (Jun):** World Cup live-scores widget
 - **v152 (Jun): Shake to Summarize / page summaries for all (English)** via Mistral Small 3.1; **Tab Groups on Android**; PDF file sharing; pinch-zoom performance overhaul
 - **v152.0.4 (Jun):** "You're protected" home message only after trackers actually blocked

Edge (Android)

1. Overview & positioning

- **Package:** `com.microsoft.emmx` (Google Play), Chromium/Blink-based (not WebView), synced release cadence with desktop Edge (~4-week major versions; v150 as of July 2026).
- **Positioning:** Microsoft's "AI browser" and full-service companion to Windows/desktop Edge — pitched as "the browser with Copilot built in." Differentiators vs. Chrome on Android: built-in Adblock Plus, mobile extension support, Microsoft Rewards earning, deep Copilot/Bing/MSN integration, and cross-device continuity with Windows.
- **Strategic direction (2025–2026):** aggressive consolidation around Copilot. In May 2026 Microsoft **retired the standalone "Copilot Mode"** and folded its AI features into the default browser UI on desktop and mobile. Simultaneously it has been **pruning legacy features** (Collections and Sidebar deprecated around Edge 149; **Drop retirement announced June 2026**) to "make room for Copilot."
- Target user: Microsoft-account households, Windows users, Rewards point earners, and enterprise (extensive Intune MAM policy surface on Android — release notes are dominated by enterprise policies like `ExtensionInstallForcelist`, `DownloadBlockedForFileTypes`, `CopilotNewTabPageEnabled`).
- Market share on Android remains small single-digits; the app is heavily promoted via Windows, Bing, and Rewards funnels.

2. Feature inventory by category

Navigation & address bar

- Combined search/URL omnibox with Bing default; search suggestions, site suggestions, Copilot suggestions.
- **Top or bottom address bar** — user-selectable position (bottom option shipped early 2025, long-requested).
- Voice search microphone in the address-bar keyboard panel.
- Camera icon: **visual search (Bing Lens-style) + QR/barcode scanning** from the address bar; QR scanner also in main menu (v150 moved scanning to Edge's built-in camera UI).
- Copilot **Summary notification chip in the address bar** (page-summary entry point, 2026).
- Swipe horizontally on the address bar to switch tabs (Chrome-style; extended to bottom omnibox).

Tabs & tab management

- Grid tab switcher with normal / InPrivate panes; "+" for new tab, close-all option.
- **Tab groups on Android** (rolled out from Canary flag `Edge Tab auto-grouping` to stable): auto-grouping, manual add/remove, rename group, ungroup, hide/collapse groups — near desktop parity.
- Recently closed tabs / reopen; switch-tabs via address-bar swipe.
- Copilot can **reason across all open tabs** (compare, summarize, answer questions) — 2026.
- "Continue on PC"-style cross-device tab handoff via Microsoft account sync (send tabs between phone and desktop).

Home / New-tab page (feed)

- NTP: search box, wallpaper, quick-link shortcut tiles (top sites + user pins; enterprise `ManagedTopSites` / `ManagedNtpShortcuts`), weather card, and an **MSN (Microsoft Start) news feed** below the fold.
- Three layout presets via NTP gear icon: **Focused / Inspirational (Bing daily image) / Informational (full news dashboard)**, plus custom; feed can be turned off.
- **Copilot New Tab experience** (2026): Copilot prompt card / Copilot-centric NTP rolled from desktop to mobile; a "Copilot New Tab Card" surfaces prompts and continuation of prior tasks (policy-controllable via `CopilotNewTabPageEnabled`).
- Rewards and shopping promos appear on the NTP in some markets.

Search & discovery

- Bing default (switchable to Google, Yahoo, DuckDuckGo, etc. in Settings → General → default search engine).
- Visual search from camera icon; voice search; QR scan.
- Text-selection "search in sidebar / web search" context action.
- Bing Rewards-earning searches count on mobile Edge.

- Journeys (see AI) doubles as history-based re-discovery.

Privacy & security (tracking prevention tiers)

- **Tracking Prevention with three tiers — Basic / Balanced (default) / Strict** — same Trust Protection List engine as desktop, under Settings → Privacy and security.
- InPrivate browsing mode (with option to lock/require authentication on some builds).
- Microsoft Defender **SmartScreen** phishing/malware protection.
- Password Manager with autofill, password monitor/leak alerts tied to Microsoft account sync.
- Clear browsing data on exit options; site permissions per-site; cookie controls; "Do Not Track" toggle.
- Enterprise: Intune MAM containerization, conditional access, `IncognitoModeAvailability`, allowed/blocked URL policies — the deepest MDM surface of any Android browser.
- Criticism noted in privacy audits: heavy telemetry defaults; personalization + ads tied to Microsoft account (see §5).

Ad/content blocking (built-in Adblock Plus)

- **Built-in Adblock Plus content blocker** — a bundled ABP engine, toggleable ("Block ads"), with Acceptable Ads implications; a genuine differentiator vs. Chrome.
- The "Block Ads" toggle was **hidden/relocated during the 2025 settings redesign** (Settings → Privacy and security → Content blockers on current builds); Microsoft says the engine wasn't removed, just re-homed — caused user confusion.
- Ad blocking can now also be supplemented with **real ad-block extensions** (uBlock Origin Lite, AdGuard, Ghostery) via mobile extension support — AdGuard was among the first partners.
- Standard pop-up blocker.

Downloads & media

- Download manager with in-app list, notifications, "Ask where to save," background downloads; PDF download-to-OneDrive integration.
- Built-in **PDF reader** (with Copilot PDF summarization); PDF print.
- **Picture-in-Picture overlay** for supported video pages (policy `PictureInPictureOverlayEnabled`, 2026).
- Copilot **Video Summary** for videos incl. YouTube (2026).
- Files can be shared to OneDrive; media autoplay controls per site.

Reading & content (immersive reader, read aloud, translate)

- **Immersive Reader** reading view (icon in address bar on compatible pages): decluttered text, text size/spacing, themes, grammar tools.
- **Read Aloud** with natural neural voices, works on pages and PDFs; controllable by `ReadAloudEnabled` policy (2026) — works with screen off in some builds.
- **Microsoft Translator** built in: full-page translate prompts and on-demand translation of selection or page.
- Copilot page **Summarize** (address-bar chip / menu action).

Customization & appearance

- Light/dark/system theme; NTP wallpaper/background images; NTP layout presets.
- **Address bar position top/bottom.**
- **Customizable main menu:** drag-and-drop reorder of actions ("Change menu"/edit button); grid vs. list presentation (layout varies by rollout and system font size).
- Text scaling/page zoom; homepage button configuration.
- Redesigned toolbar (2025–2026): re-arranged bottom bar with Share button replacing "Continue on PC," centered ellipsis menu.

Sync & account (Drop, Collections, cross-device)

- Microsoft-account sign-in (MSA + Entra ID work profiles, multi-profile with easy identity switching).
- **Sync:** favorites, passwords, addresses/payment, open tabs, history, Collections (toggleable per data type); "Reset sync server data" option added 2025.
- **Collections on mobile:** save pages/notes to synced Collections via menu → "Add to Collections" — **but Collections is being deprecated** (removed from desktop around Edge 149; mobile long-term future doubtful).
- **Drop:** send-to-self files/notes thread synced via OneDrive, available in mobile menu — **officially being retired (announced June 2026)**; text notes will be deleted, files remain in OneDrive.

- Cross-device: send tab to devices, shared password autofill, continue browsing on PC.

AI features (Copilot)

- **Copilot chat built in** (dedicated button/menu entry; formerly Bing Chat): Q&A, composition, image understanding, GPT-class models; free tier plus Copilot Pro benefits.
- **Copilot Mode retired May 2026** — AI features now integrated throughout the default UI.
- **Multi-tab reasoning**: Copilot analyzes/compares/summarizes across open tabs; can continue tasks started on desktop.
- **Journeys on mobile** (2026): organizes browsing history into topic-based cards with summaries and suggested next steps.
- **Copilot Vision & Voice on mobile** (2026): share your screen and talk through what you're looking at, hands-free.
- **Copilot long-term memory** across sessions (2026).
- Page/PDF/video **summarization**; Copilot Summary address-bar notifications; Copilot prompt card on NTP.
- New 2026 experiments: turn open tabs into **podcasts**, generate **quizzes** from webpages.
- Enterprise-grade "AAD Copilot" (work-account Copilot with commercial data protection) on Android.

Gestures & shortcuts

- Swipe address bar left/right to switch tabs (works with bottom omnibox).
- Swipe down from address bar for tab management/switcher (build-dependent).
- Pull-to-refresh on pages.
- Long-press links/images for context menus; long-press tab-switcher and toolbar buttons for quick actions (e.g., new InPrivate tab).
- Known annoyance: tab-switch swipe order ignores tab-group visual order.

Data saving & performance

- No classic proxy "data saver" mode (the old Turbo/data-compression died with legacy Edge); efficiency comes from ad/tracker blocking reducing payloads.
- Startup/parallel tab-load tuning (`AntiParallelOnStartupEnabled`), background tab throttling inherited from Chromium.
- Criticized as heavier (APK size, RAM, battery) than Chrome/Firefox due to bundled services (Rewards, shopping, Copilot, MSN feed).

Extensions/add-ons (mobile extension support!)

- **Official extension support on Android** — flagship differentiator. Piloted late 2024 (~20 curated add-ons: uBlock, Dark Reader, Tampermonkey, AdGuard, Bitwarden-class), **official global rollout March 2025**, on by default for Android 11+ by 2026.
- Dedicated mobile **Edge Add-ons store**; curated catalog (~87 approved of 2,400+ submitted as of March 2026) — approval requires mobile-safe UI, no battery-draining background scripts, no Android system-API access.
- Late-2025 beta expanded compatibility toward "virtually any" desktop-class Edge/Chromium extension (thousands installable), though not full parity — some render poorly on mobile.
- Enterprise `ExtensionInstallForceList` policy for silent extension deployment on Android (v150, July 2026).

Accessibility

- Read Aloud + Immersive Reader (core accessibility pair).
- Text scaling/zoom overriding site restrictions; respects Android font-size/bold accessibility settings (which even flips menu grid→list layout).
- TalkBack screen-reader support; high-contrast/dark theme; Copilot Voice/Vision as hands-free assistive interaction.

Shopping/Rewards/extras

- **Microsoft Rewards integrated**: join/track points in Settings → My profile → Microsoft Rewards; earn on Bing searches in mobile Edge; redeem for gift cards etc. (select markets).
- Shopping: price comparison, **price history and price-drop alerts**, cashback in supported regions; native Bing-powered coupon experience was ported to Android — **the standalone Coupons feature was retired (2025)** in favor of cashback/price tools.
- Extras: built-in games (Surf game — now policy-gated `AllowSurfGame`), Bing daily wallpapers, MSN content ecosystem, Web Capture/screenshot tool, share sheet with QR-code page sharing, "Send to devices."

3. UI/UX structure & feature placement

- **Address bar position and contents:** Top by default, **user-movable to bottom** (Settings → appearance/address bar). Contains: site security/lock (page info incl. SSL cert info), URL/search field, refresh; focused state shows voice mic + camera (visual search/QR) above keyboard; Copilot Summary chip can appear in the bar. Swiping the bar switches tabs.
- **Bottom toolbar layout:** Persistent bottom bar (Edge's signature vs. Chrome) — typically **5 slots: Back · Forward (or Home) · Tab switcher (count badge) · Copilot (center-ish, replaced older hub/"Continue on PC"; a Share button appeared in the 2025 redesign) · "... ellipsis menu**. The 2025–2026 redesign moved the ellipsis toward center and swapped Continue-on-PC for Share. The toolbar itself is only lightly customizable; deep customization lives in the menu sheet.
- **Tab switcher design:** Full-screen grid of tab thumbnails; toggle between normal and **InPrivate** spaces; tab-group cards with rename/ungroup/hide via group ellipsis menu; + new tab, close-all, and overflow menu.
- **Main menu (customizable bottom-sheet grid):** Opens from "... as a sheet — historically a **paged icon grid**, being A/B-migrated to a **reorderable list** ("Change menu" enables drag-and-drop; grid vs. list also depends on system font size and rollout). Typical items: New tab · New InPrivate tab · Add to favorites · Favorites · History · Downloads · Collections · Copilot · Share · Find on page · Translate · Read aloud · Desktop site toggle · Add to phone (home screen) · Drop (being retired) · QR scanner/scan code · Web capture · Night/dark mode · Rewards · Settings · Exit. Top row of the sheet shows account/profile switcher.
- **Settings screen — top-level sections (post-2025 redesigned settings, approximate order):** Profile/account card (sign-in, sync, **Microsoft Rewards** under My profile) → **General** (default search engine, homepage, address bar position, default browser) → **Accounts/Sync** → **Appearance** (theme, menu/toolbar) → **New tab page** → **Privacy and security** (tracking prevention tiers, content blockers/Block ads, clear data, InPrivate, Do Not Track) → **Passwords/Autofill** (payments, addresses) → **Site permissions** → **Downloads** → **Languages/Translate** → **Accessibility** → **Notifications** → **Copilot/AI settings** → **About**. (Exact order shifts across the ongoing settings-UI migration — a documented source of user complaints.)
- **Long-press / context menus:** Links: open in new tab / new InPrivate tab / open in group, copy link, share, download link, preview. Images: open, save, share, **visual search with the image**, copy. Text selection: copy, share, web search, translate, Copilot/ask. Address bar long-press: paste/copy URL. Home-screen app icon long-press: shortcuts (new tab, InPrivate tab, search).
- **Gestures supported:** pull-to-refresh; address-bar horizontal swipe = tab switch; swipe down from bar = tab UI (build-dependent); pinch zoom; back-gesture integration with Android navigation; swipe tabs closed in switcher.
- **First-run/onboarding flow:** Splash → Microsoft account sign-in prompt (auto-suggests device MSA; skippable) → sync opt-in → data/telemetry + personalization consent (region-dependent) → **set-as-default-browser prompt** (Android role manager sheet) → NTP feed/layout intro; 2026 builds add a **Copilot FRE (first-run experience) screen for MSA users** introducing AI features. Recurrent nudges thereafter: sign-in banners, default-browser prompts, Rewards/Copilot promos — a common annoyance theme.

4. Standout features (what Edge does best)

- **Mobile extension support** — the only mainstream Chromium browser on Android with an official, growing extension store (uBlock Origin Lite, Dark Reader, AdGuard, Tampermonkey...), heading toward desktop-catalog compatibility.
- **Built-in Adblock Plus** ad blocking out of the box — no setup, unlike Chrome.
- **Deepest mobile AI integration of any major browser:** Copilot multi-tab reasoning, page/PDF/video summarization, Journeys, Vision + Voice screen-sharing assistant, long-term memory, tabs-to-podcast/quiz experiments.
- **Microsoft Rewards earning while browsing/searching** — effectively pays users; unique among browsers.
- **Cross-device continuity with Windows:** tab/password/favorites/Collections sync, send-to-PC, shared Copilot context.
- **Bottom address bar option + persistent bottom toolbar** — strong one-handed ergonomics.
- **Reading stack:** Immersive Reader + neural Read Aloud + full-page Translate is best-in-class on mobile.
- **Enterprise manageability:** unmatched Intune/MAM policy coverage on Android (forced extensions, download blocking, managed NTP, Copilot controls).
- Shopping tools (price history, price alerts, cashback) built in.

5. Weaknesses & common criticisms

- **Bloat/clutter:** Rewards, shopping, MSN feed, Copilot everywhere — reviewers call it overwhelming vs. minimalist rivals; "Microsoft services detract from focused browsing."
- **Aggressive promotion:** persistent sign-in, default-browser, Bing/Copilot and Rewards nudges; "Stay in Microsoft Edge" interstitials hijacking external-app redirects irritate users and break some OAuth flows.
- **Feature churn/removals:** Collections, Sidebar, Coupons, and now **Drop** all deprecated in ~18 months to clear room for Copilot — erodes trust in investing in Edge features; Drop text notes get deleted.
- **Settings redesign confusion:** options (notably the Block Ads toggle) moved or hidden mid-2025; grid-vs-list menu inconsistency across identical devices.

- **Privacy reputation:** heavy telemetry defaults, Bing/MSN data coupling, ABP "Acceptable Ads" model; privacy guides recommend extensive toggling.
- **Performance/resource footprint:** heavier RAM/battery/APK than Chrome; users report the mobile engine trailing desktop Chromium in features/security cadence.
- **Extension catalog still curated/small** in the official store (~87 approved as of March 2026), with mobile-rendering issues for desktop-class extensions in the expanded beta.
- **Tab UX rough edges:** swipe-to-switch ignores tab-group ordering; tab opening direction inconsistencies.
- No true data-saver mode; AI features gated by Microsoft account (and best experience by Copilot Pro).

6. Recent additions (2024–2026)

- **Late 2024:** extension support pilot on Android (curated ~20 add-ons); Copilot (Bing Chat successor) embedded in bottom bar; tab auto-grouping in Canary.
- **Feb–Mar 2025: bottom address bar option** ships (Feb 2025); **official global extension rollout** (Mar 2025); "Reset sync server data"; toolbar redesign begins (Share replaces Continue-on-PC, centered ellipsis); customizable/reorderable actions menu experiments; native Bing coupon experience replaces Honey-style coupons (then Coupons retired outright).
- **Mid–late 2025:** settings UI overhaul (Block Ads toggle relocated); tab groups mature toward stable (rename/ungroup/hide); Copilot PDF summarization and YouTube/video integration; October 2025 beta expands extension compatibility toward the full desktop catalog.
- **Early 2026:** Copilot Mode FRE for MSA users; Copilot prompt card on NTP; Copilot Summary notification in address bar (v145, Mar 2026); multi-tab summary + video summary polish (v148, May 2026); `CopilotNewTabPageEnabled`, `ReadAloudEnabled`, `EdgeEDropEnabled`, PiP overlay policies; extensions default-on for Android 11+.
- **May 2026: Copilot Mode retired; AI integrated into default UI** — multi-tab reasoning, **Journeys on mobile**, **Vision + Voice on mobile**, long-term memory, tabs-to-podcast, quiz generation; desktop-style new-tab page to mobile (Edge blog, May 13, 2026).
- **June–July 2026: Drop retirement announced** (following Collections/Sidebar deprecation ~Edge 149); v150 (July 2026): `ExtensionInstallForcelist` on Android, QR scanning via built-in Edge camera, Copilot Summary/New Tab Card refinements.

Samsung Internet (Android)

(Rebranded to "Samsung Browser" starting with the Galaxy S26 / One UI 8.5 in early 2026; the rebrand is rolling back to older Galaxy devices via One UI 8 updates. Sources still use both names — this report treats them as the same product.)

1. Overview & positioning

- **What it is:** Samsung's first-party Chromium/Blink browser, default on all Galaxy phones and tablets. Debuted 2012 (Galaxy S3, AOSP-based), Chromium-based since 2013 (Galaxy S4), published on Google Play since 2017 so it installs on any Android 10+ device (non-Samsung included, with some Galaxy-only feature gaps).
- **Scale:** ~2.3–2.8% of *global* mobile browser share as of early-to-mid 2026 (StatCounter ~2.81% April 2026) — 3rd-largest mobile browser after Chrome and Safari and the **largest non-default mobile browser**; 1B+ Play Store downloads. Peak share was ~7.4% (Jan 2019); the decline reflects Chrome-as-default pressure, not feature regression.
- **Positioning:** the "power-user Chrome alternative that ships by default" — identity pillars: (1) one-handed ergonomics (bottom bar, customizable toolbar), (2) privacy defaults (Smart Anti-Tracking on by default, biometric Secret mode), (3) content-blocker plugin ecosystem Chrome lacks, (4) Galaxy ecosystem tie-ins (Samsung Pass/Knox, Galaxy AI Browsing Assist, cross-device sync).
- **Version cadence:** roughly monthly-to-quarterly major versions; as of June 2026 the Android stable is **v30.0.0.67**. Also now on **Windows** ("Samsung Browser for Windows," beta Oct 30 2025, GA 2026) with full Chrome-extension support and Samsung Account sync.
- **Reputation:** widely used as a UX benchmark for one-hand mobile browsing; reviewers repeatedly note "Galaxy users who consciously switch to Samsung Internet rarely switch back."

2. Feature inventory by category

Navigation & address bar

- **Bottom or top address bar** — user-selectable position; bottom placement predates Chrome's equivalent by years.
- **Combined or separated tab/address bar layout** — "Together" (one line) or "Separate" via Settings > Layout and menu.
- **Swipe-on-address-bar tab switching** — swipe left/right on the address/bottom bar to move between tabs.
- **Settings search** — search box inside Settings to find any option (added v27).
- **QR code scanner** — launchable from the toolbar/menu (assignable button).
- **Toolbar blur effect** — One UI 8.5/9 adds a translucent blurred toolbar background (floating, layered look).

Tabs & tab management

- **Tab groups** — create, name, rename, reorder, share (shares all URLs in group), and close groups.
- **Tab group sync** — tab groups sync across devices via Samsung account (since v23, Oct 2023).
- **Three tab-switcher layouts** — List, Stack, or Grid view ("View as" option).
- **Tab locking** — lock individual tabs against accidental closure.
- **Auto-close unused tabs** — automatically closes tabs untouched for a set period (2025 addition).
- **Reopen closed tabs / recently closed list.**
- **Tab bar on large screens** — desktop-style visible tab strip on tablets/foldables (optional on phones).
- **Multiple windows** — One UI 9 (late 2026) adds true simultaneous multi-window browser instances, even on regular phones.

Home / New-tab page (Quick access)

- **Quick access grid** — user-editable icon grid of favorite sites; add/remove/reorder; can be set as homepage, or homepage can be Current page / Custom URL / blank.
- **News feed on Quick access** (region-dependent) — personalizable by language, location, and topic categories; can be fully disabled ("None").
- **Home screen shortcuts** — pin any site to the Android launcher (improved in v26).
- **Redesigned home screen** — refreshed layout shipped during 2025.

Search & discovery

- **Switchable search engines** — Google (default), Bing, Yahoo, DuckDuckGo, Yandex, and regional options.

- **Search suggestions & unified URL/search field.**
- **In-page search (Find on page)** via main menu.
- **Video history** — dedicated history of videos watched in-browser, separate from page history.
- **History sorting options** and longer history retention (expanded 2023–2025).

Privacy & security (Secret mode, Smart anti-tracking)

- **Secret mode** — private browsing that saves no history/cookies/search records; lockable with **password + biometrics (fingerprint or face)** — a differentiator vs. Chrome Incognito's simple lock.
- **Secret mode options** — allow/deny screenshots in Secret mode (screenshots allowed since v26), option to stay in Secret mode after closing browser, separate anti-tracking strength for Secret mode.
- **Smart Anti-Tracking** — AI-driven intelligent tracking prevention, **on by default**; blocks cross-site trackers, deletes tracking cookies/web traces, resists circumvention techniques; adjustable protection levels (Basic/Strong); includes automatic CAPTCHA-verification assistance.
- **Privacy Dashboard** — real-time count/report of blocked trackers and privacy status.
- **Per-site permissions** — camera, mic, location controls per site.
- **Warn-before-malicious-sites / anti-phishing protections**; passkey support (WebAuthn) with device biometrics.
- **End-to-end encryption for synced data** (v27, late 2024).
- **Quick Exit button** (assignable) — one tap closes everything, optionally wiping history and cookies.

Ad/content blocking (third-party content blocker plugins)

- **Content-blocker extension API** — Samsung Internet pioneered (2016) an installable content-blocker plugin system on Android; blockers install as normal Android apps from Play Store/Galaxy Store, then activate under Menu > Settings/Ad blockers.
- **Multiple blockers simultaneously** — users can enable several at once.
- **Ecosystem** — AdGuard Content Blocker (20+ filter lists, custom rules, no root), Adblock Plus for Samsung Internet, Adblock for Samsung, Crystal, Unicorn, Disconnect, etc. The browser surfaces a curated recommendation list in-settings.
- **Limitation** — content blockers are declarative filter lists (Safari-style), not full extensions; no element zapping or scripting.

Downloads & media (video assistant, save video)

- **Video Assistant** — floating control on web videos offering: pop-up (picture-in-picture) player, one-tap full screen, and "view on TV" casting; gesture controls inside player (swipe for brightness/volume, double-tap seek). Toggle at Settings > Useful features > Video Assistant.
- **Download manager** — pause/resume, download history, choose storage location.
- **Save page / save image / save link** via long-press context menu.
- **Video history** menu for revisiting watched videos.

Reading & content (reader mode, translate, text scaling)

- **Reader mode** — declutters articles; adjustable font and font size (limited background-color choices — a noted weakness).
- **Translate** — full-webpage translation; engine improved 2026 with more accurate output across 20+ languages; deep integration with Galaxy AI on-device translation that even replaces text in menus and video overlays in-place.
- **Text scaling** — Settings > Accessibility: text size slider (can override site-forced sizing) plus webpage zoom control independent of system font size.
- **Read Highlight Aloud** — TTS reads selected/highlighted text.

Customization & appearance (customizable toolbar/menu, dark mode for web)

- **Customizable bottom toolbar** — up to 7 slots, filled by drag-and-drop from the full function catalog (back/forward, home, tabs, bookmarks, downloads, share, dark mode, Quick Exit, QR scanner, desktop site, add-ons, Secret mode toggle, etc.). Introduced v10.2 (2019); still nearly unique among mainstream mobile browsers.
- **Customizable main menu grid** — the ≡ menu's icon grid is also fully re-arrangeable via the same drag-and-drop "Customize menu" screen (Settings > Layout and menu).
- **Address bar position** — top or bottom.
- **Dark mode for web content** — force-darkens all websites (algorithmic transformation), separate from UI dark theme; toggleable from the menu; known to occasionally mangle link colors/contrast.
- **High contrast mode** — accessibility-oriented white-on-dark rendering.
- **One UI 8.5/9 visual refresh** — blur/translucency, floating toolbar elements, new app icon and "Samsung Browser" branding.

Sync & account (Samsung Cloud)

- **Samsung account sync** — bookmarks, saved pages, open tabs, tab groups, history, and (via Samsung Pass) passwords/passkeys across Galaxy phones, tablets, and now Windows PCs.
- **E2E-encrypted sync** (since v27).
- **Cross-device resume** — Windows/mobile "continue where you left off" prompts.
- **No Google-account sync** — sync universe is Samsung-account-only.

AI features (Galaxy AI browsing assist)

- **Browsing Assist** (One UI 6.1+, Android 14+, Galaxy AI devices): **Summarize** — on-device AI page summaries with concise/detailed levels, plus translation of summaries; **Translate** — full-page on-device translation (20+ languages) that in-place replaces text including menus, ads, and text embedded in video.
- **Intelligent Summarization** improvements through 2025–2026 for long-form articles.
- **Ask AI** (March 2026, One UI 8.5, US/Korea first) — contextual Q&A about the page you're viewing. Powered by Perplexity APIs.
- **Ask Anything** (One UI 9, late 2026) — Ask AI renamed/expanded: answers questions using page context *and* browsing history.
- **Notes PDF** integration (One UI 9, in testing) — send page content into Samsung Notes as PDF.
- Requires Samsung Account sign-in; some features need network despite on-device models.

Gestures & shortcuts

- **Swipe address bar / bottom bar left-right** — previous/next tab (phones; tablets use tab strip instead).
- **Pull-to-refresh**.
- **Video player gestures** — vertical swipes for brightness (left) / volume (right), double-tap seek.
- **Edge-swipe back/forward** — follows system gesture nav.
- **Long-press toolbar icons** for secondary actions (e.g., long-press tabs button for new tab/close tab options).

Data saving & performance

- **No dedicated data-saver proxy** — content blockers are the de-facto data saver.
- **Auto-close unused tabs** reduces memory footprint.
- **AMOLED battery benefit** from forced dark mode is marketed as a power feature.
- **Chromium-current engine** (v30 tracks recent Chromium) though historically it lags Chrome's engine version by several releases — a recurring criticism.

Extensions/add-ons

- **Two-tier model:** (1) open **content-blocker API** (Play Store apps, any Android device); (2) general **"Add-ons"** — translation tools, shopping assistants, security tools — distributed via **Galaxy Store** under "Samsung Internet Extensions," effectively Galaxy-device-only and a **closed beta** (only approved developers can publish).
- Legacy built-in add-ons: **Web contents provider / Amazon shopping assistant, Translator extension**.
- **Windows version supports full Chrome Web Store extensions** — mobile does not.

Accessibility

- **Text size slider + webpage zoom** independent of system settings; "force zoom" on sites that block pinch-zoom.
- **High contrast mode** (Settings > Accessibility).
- **Reader mode** for clutter-free reading.
- **Read Highlight Aloud** TTS.
- Works with TalkBack; large-touch-target bottom UI is itself an accessibility win.

Unique extras (Knox, Samsung Pass, etc.)

- **Samsung Pass integration** — biometric web login and autofill (IDs/passwords/OTP/addresses/cards); credentials stored in the **Knox** hardware-isolated vault; biometrics never leave the device; passkeys stored in Samsung Pass.
- **Knox-hardened browsing** — Knox platform integration; managed-configuration support for enterprise.
- **Biometric web authentication (WebAuthn/FIDO)** — fingerprint/face sign-in to websites.
- **DeX support** — desktop-class windowed browsing in Samsung DeX.

- **Quick Exit** panic button with data wipe.
- **Continue on PC** — cross-device handoff with Samsung Browser for Windows.

3. UI/UX structure & feature placement

- **Address bar + bottom toolbar layout & customizability:** Default layout is address bar (top or bottom, user's choice) plus a persistent bottom toolbar. The toolbar holds **up to 7 buttons** chosen by the user: Settings > **Layout and menu** > **Customize menu** opens a two-pane editor — the current toolbar/menu on top, the catalog of all available functions below — and users **drag and drop any function into any slot and reorder freely**. Items that don't fit on the toolbar live in the overflow menu grid, which is re-orderable by the same drag-and-drop. Tab bar and address bar can be merged into one row ("Together") or split ("Separate"). This is the deepest toolbar customization of any mainstream mobile browser and the core of its one-handed pitch.
- **Tab switcher design:** Full-screen switcher opened from the toolbar tabs button; three selectable presentations — **List, Stack (rolodex), Grid** — plus tab-group sections with named, colored group headers; bottom row offers new tab / Secret mode toggle; three-dot menu inside the switcher exposes Group tabs, Edit, Share group, Close all. Tablets/foldables get a desktop-style persistent tab strip instead.
- **Main menu (the grid sheet):** The ≡ (or :) button opens a bottom-sheet **icon grid** (two+ rows) containing roughly: Bookmarks, Downloads, History, Saved pages, Share, Find on page, Desktop site, Add-ons/Ad blockers, Text size, Dark mode, Translate, Print, Zoom, Video history, Settings, and (recent) AI actions. The grid contents and order are **user-customizable**; anything can be promoted to the toolbar or demoted to the sheet.
- **Settings: top-level sections in order (approximate):** Samsung account/sync → **Layout and menu** → **Homepage** → Search engine/Address bar → **Tabs** → **Privacy and security** (incl. Secret mode settings, Smart Anti-Tracking, Privacy dashboard) → Ad blockers / **Add-ons** → Sites and downloads → **Useful features** (Video assistant, dark mode on websites, open links in other apps, etc.) → **Accessibility** → Labs/experimental → About. A settings **search bar** (v27+) sits on top.
- **Long-press / context menus:** Long-press link → open in new tab / open in Secret tab / open in group, copy, share, save link, download; long-press image → save, copy, share, search-by-image; long-press text selection → copy, share, web search, **Translate**, (AI actions on Galaxy AI devices); long-press toolbar buttons for secondary actions; long-press Quick access icons to edit/remove.
- **Gestures supported:** address-bar/toolbar horizontal swipe = tab switching; pull-to-refresh; system edge-swipe back; video-player brightness/volume swipes and double-tap seek; pinch zoom (with force-enable option).
- **One-handed-use design choices:** everything critical is reachable at the bottom — bottom address bar option, bottom toolbar, bottom-sheet main menu that opens above the thumb, bottom-anchored tab switcher controls, swipe-on-bar tab switching so no reach to top corners. This bottom-first architecture (shipped years before Chrome tried it) is why the browser is a one-hand UX benchmark.

4. Standout features (what Samsung Internet does best)

- **Toolbar + menu customization** — no mainstream mobile rival lets users drag-and-drop reorder up to 7 toolbar buttons *and* the whole overflow menu.
- **Secret mode with biometric lock** — password/fingerprint/face gate on private browsing; more robust than Chrome's Incognito lock, with its own anti-tracking level and persistence options.
- **Content-blocker plugin ecosystem** — the only major Android browser with a third-party ad-block extension API (AdGuard, Adblock Plus, Crystal, Unicorn...), running since 2016.
- **Video Assistant** — pop-up player, casting, and in-player gestures on arbitrary web video.
- **Smart Anti-Tracking on by default + Privacy Dashboard** — strongest default privacy posture among default OEM browsers.
- **Galaxy AI Browsing Assist** — on-device summarize + seamless in-place full-page translation, now expanding to Ask AI / Ask Anything.
- **Bottom-first, one-hand ergonomics** — the industry reference for reachable mobile browser UI.
- **Tab management depth** — groups (synced), lock, three switcher layouts, auto-close stale tabs, and (One UI 9) true multi-window.
- **Samsung Pass/Knox biometric autofill & passkeys** — hardware-vault credentials with biometric web login.

5. Weaknesses & common criticisms

- **Galaxy-ecosystem lock-in:** best features require a Samsung account and often Galaxy hardware; on non-Samsung Androids it's a diminished product.
- **Extension ecosystem is shallow and gated:** mobile add-ons are a closed beta on Galaxy Store; content blockers are filter-list-only — no uBlock-style full extensions on mobile.
- **Engine lag:** Chromium base historically trails Chrome by multiple versions, raising security-patch and web-compat concerns.
- **No Google sync / weak cross-ecosystem sync.**
- **Forced dark mode breaks pages:** algorithmic darkening can produce unreadable link colors and contrast failures.

- **Video Assistant bubbles obscure page UI:** floating buttons famously cover sites' own controls.
- **News feed / promotional clutter:** region-dependent feed and Samsung-services promotion feel like bloat.
- **Reader mode is basic:** limited theming/background options.
- **Update pacing & feature fragmentation:** flagship features debut on newest One UI first, fragmenting the experience.
- **Declining market share:** from 7.4% (2019) to ~2.3–2.8% (2026), squeezed by Chrome defaults.

6. Recent additions (2024–2026)

- **Early 2024 (v24–25, One UI 6.1):** Galaxy AI **Browsing Assist** — on-device Summarize and full-page Translate.
- **Mid 2024 (v26):** screenshots allowed in Secret mode; home-screen site shortcuts improvements.
- **Late 2024 (v27):** **end-to-end encryption for synced data; settings search;** sync scope expansion.
- **2025 (v28–29):** new tab-management features; **auto-close unused tabs;** history sorting; redesigned home screen.
- **Oct 2025: Samsung Internet for Windows PC beta** — Chromium, Chrome extensions, Samsung Pass + Windows Hello autofill, cross-device sync, Browsing Assist AI; GA and global expansion through 2026.
- **Mar 2026 (One UI 8.5 / Galaxy S26 launch):** rebrand to **"Samsung Browser"** with new icon/identity; **Ask AI** contextual page Q&A (US/Korea first); improved translation engine (20+ languages).
- **Mid 2026 (One UI 9 previews/beta):** **multiple simultaneous browser windows; Ask Anything; toolbar blur/translucent redesign; Notes PDF** export integration; "Cross Device Resume" flag spotted.
- **Ongoing 2026:** v30.x on Android; expansion of Samsung Browser beyond Galaxy devices.

Brave (Android)

1. Overview & positioning

- **Positioning:** Privacy-first Chromium browser whose core pitch is "block everything by default" — ads, trackers, fingerprinting — with an opt-in attention economy (BAT rewards) and a growing privacy-preserving AI stack (Leo, Brave Search). Marketed as "3x faster than Chrome" because blocked ads/trackers mean fewer network requests and less CPU/battery/data.
- **Engine:** Chromium (Blink). Tracks Chromium closely — as of mid-2026 stable Android is ~1.91–1.92 on Chromium 149–150.
- **Scale:** ~65M+ MAU across platforms per 2026 comparisons; consistently ranked the top privacy pick among Android browsers in 2026 roundups.
- **Business model:** Free browser; monetized via opt-in Brave Ads/Rewards (BAT), Brave Search ads, and paid subscriptions: Leo Premium (\$14.99/mo), Brave Firewall+VPN (\$9.99/mo), Search Premium, and — new in 2026 — **Brave Origin**, a \$59.99 one-time-purchase "de-bloated" browser variant with the crypto/AI/news extras stripped out.

2. Feature inventory by category

Navigation & address bar

- Combined omnibox (URL + search + suggestions) with "Ask Leo" entry — typed queries can be routed to Leo AI directly from the address bar.
- Address bar position: **top by default, bottom optional** — toggle in Appearance settings or long-press the bar.
- Shields (lion) icon lives inside the address bar; tap for the per-site Shields panel.
- **Custom toolbar shortcut** (v1.85+): a configurable toolbar button giving one-tap access to a chosen feature.
- Swipe left/right on the toolbar (top or bottom) to switch tabs.
- De-AMP (bypass Google AMP pages to publisher's real URL), debouncing (skip known tracking redirect hops), tracking-parameter stripping on navigation.

Tabs & tab management

- Chromium-style grid tab switcher with separate **Private tabs** pane; tab groups supported.
- Recent tabs / synced tabs from other chain devices.
- Fixes in 2026 for foldables and dynamic-theme tab-switcher colors.

Home / New-tab page

- NTP shows **Privacy Stats**: lifetime ads/trackers blocked, bandwidth saved, time saved (headline Brave differentiator).
- **Background images** and **Sponsored Images** (full-page NTP ad backgrounds; earn BAT if Rewards is on). Toggles: Settings → Display → New Tab Page.
- Top Sites row beneath stats; **pinned website shortcuts** addable to NTP since v1.86.
- **Brave News**: private, on-device-personalized news feed built into the NTP scroll; can be disabled.

Search & discovery (Brave Search)

- Brave Search is the default engine (independent index; no Google/Bing dependency); default can be changed; custom search engines supported.
- Separate default engine settable for **standard vs private tabs**.
- **Ask Brave** (Sept 2025→2026): unified search+AI chat — chat follow-ups, and a **Deep Research mode** running multi-round searches over Brave's index; free.
- **Rerank**: user-side boosting/demoting/removal of domains in search rankings.
- Web Discovery Project: opt-in, anonymous contribution of browsing data to improve the Brave Search index.

Privacy & security (Shields)

- **Shields master toggle per site** (lion icon): per-site overrides of global defaults set at Settings → Brave Shields & privacy.

- **Trackers & ads blocking: Disabled / Standard / Aggressive.** Standard = network filtering + cosmetic filtering tuned for compatibility; **Aggressive blocks everything incl. first-party ads/trackers** and hardens against filter-circumvention scripts.
- **Fingerprinting protection:** blocks/modifies fingerprintable APIs so Brave instances look alike, plus **randomizes API outputs ("farbling")** per-site-per-session so a fingerprint can't link you across sites/sessions.
- **HTTPS by default / Upgrade connections to HTTPS**, with a strict mode that blocks plain-HTTP fallbacks.
- **Script blocking:** global "Block scripts" toggle with per-site allow; blocked-script count viewable/re-enable-able from the Shields panel.
- **Cookie controls:** block third-party cookies (default) or all cookies, per-site; ephemeral/partitioned third-party storage.
- **Forgetful Browsing:** per-site (or global) "forget me when I close this site" — auto-clears cookies/storage and logs you out when last tab of a site closes.
- **Bounce-tracking defenses:** query-parameter stripping, **debouncing**, **unlinkable bouncing**.
- **Block cookie-consent banners** and notification-permission annoyances via dedicated filter lists.
- Social media blocking: Google/Facebook login-button and embed blocking toggles.
- Safe Browsing; private tabs can be **locked behind fingerprint/biometrics**.
- **No Tor on Android** — Private Window with Tor is desktop-only.

Ad/content blocking (industry benchmark — how it works)

- Native **Rust-based adblock engine** (adblock-rust) running in the network stack — not an extension — so it's faster and can't be killed by Manifest V3; the key structural advantage vs Chrome on Android.
- **Filter lists:** ships with EasyList, EasyPrivacy, uBlock Origin lists, and Brave-maintained lists; **Filter lists page** lets users enable dozens of regional/annoyance lists, **subscribe to custom filter-list URLs, and write their own custom filter rules** — full uBlock-style syntax incl. cosmetic rules.
- **Cosmetic filtering:** hides page elements matching CSS selectors from filter lists, so pages reflow cleanly.
- **Block Elements on Android (v1.78, 2025):** tap-to-hide arbitrary page elements — Brave was the **first major mobile browser** with interactive element zapping; selections saved as per-site custom cosmetic rules.
- **Scriptlet injection** (uBO-style "resource replacement") to neutralize anti-adblock and tracking scripts; procedural cosmetic filters supported.
- YouTube-specific hardening continued through v1.87+ (better ad blocking, PiP, background play).

Downloads & media

- Standard Chromium download manager, download-later/parallel downloading flags.
- **Background video/audio playback** — free, built-in (Settings → Media → Background video playback); keep YouTube etc. playing with screen off — a flagship Android feature.
- **Brave Playlist:** save video/audio into an in-browser playlist for later/**offline** playback (Android arrival lagged iOS; some bugs).
- Picture-in-picture supported.

Reading & content (Speedreader)

- **Speedreader is effectively desktop-only.** On Android it exists only behind flags and is widely reported broken; no shipped reader mode on Android as of 2026 — a recurring user complaint.
- Page translate (Brave-hosted translation) available on Android; Leo can summarize pages as a workaround.

Customization & appearance

- Theme: light/dark/system; **dynamic (Material You) theming** via flag (2026).
- **Night Mode (experimental):** forces dark rendering of web content, with occasional rendering glitches.
- Bottom vs top address bar; **customizable main menu** (v1.84 — show/hide/reorder menu items); custom toolbar shortcut (v1.85).
- NTP background/sponsored image toggles, Brave News on/off, stats widgets on/off.
- Improved large-screen UI for tablets/foldables (2026).

Sync & account (Brave Sync chain)

- **No account, ever.** Sync uses a "Sync chain": a device generates a 25-word seed phrase (shown as a **QR code for phones**); all devices holding the seed form the chain.
- **Client-side (end-to-end) encryption** — Brave's servers store only ciphertext; sync categories selectable: bookmarks (default), history, passwords, autofill, open tabs, settings.

- Known criticism: history sync is partial, and community reports of sync flakiness persist.

AI features (Leo)

- **Leo on Android since v1.63:** page/video summarization, Q&A about the current page, translation, content generation — from the address bar ("Ask Leo") or menu → Leo.
- **Privacy architecture:** no account needed, chats not stored server-side or used for training, requests routed through an **anonymization proxy**; zero-retention model hosting.
- **Free tier** (2026): Llama 3.1 8B, Claude Haiku, Qwen 3 14B, and other open models. **Premium \$14.99/mo:** Claude Sonnet/Opus, DeepSeek, and more; higher rate limits.
- **Automatic Mode** (2026): Leo auto-picks the best model per query.
- Android-specific Leo (2025–26): quick-actions/voice widget, personalization + memories, @-mention open tabs, attach files/PDFs to chats.

Gestures & shortcuts

- Swipe horizontally on the toolbar/address bar to switch tabs; swipe up on bottom address bar opens the tab grid.
- Pull-to-refresh (not user-disableable — a complaint thread exists).
- Long-press address bar to reposition; long-press tab-switcher button for new tab / new private tab / close tab.
- Android app-shortcuts (long-press launcher icon): new tab, new private tab; Leo in quick settings (v1.82).

Data saving & performance

- No proxy-based "lite mode" — data savings come from blocking: NTP stats quantify **bandwidth saved**.
- Benchmarks: ad-heavy pages 2–3x faster than Chrome, ~44% lower RAM in 2026 head-to-heads; native Rust blocking engine keeps overhead minimal.

Rewards/Web3 (BAT, wallet)

- **Brave Rewards (opt-in):** view privacy-preserving Brave Ads → earn BAT; on-device ad matching, no profile leaves the phone.
- **Rewards 3.0 (Q1 2025):** redesigned UX; **self-custody payouts** via Brave Wallet Solana address (v1.81).
- **BAT Roadmap 3.0 "On-Chain Era" (2026):** BAT ecosystem moving on-chain; BAT usable for Brave premium products.
- **Brave Wallet:** built-in multi-chain crypto wallet — Ethereum/EVM, Solana, Bitcoin; buy/send/swap, NFTs, dApp connections.
- Creator tipping to verified sites/channels.

Accessibility

- Inherits Chromium accessibility: TalkBack support, page zoom with force-enable zoom, text scaling.
- Dedicated **Accessibility** section in settings; night/dark modes assist low-vision users; known gap: no reader mode on Android hurts accessibility use-cases.

Unique extras

- **Tor Private Windows: desktop-only.**
- **Secure DNS (DNS-over-HTTPS)** configurable on Android.
- **Brave Firewall + VPN** (\$9.99/mo, 10 devices): whole-device VPN + tracker-blocking firewall; **Smart Proxy Routing** (2025); kill switch not on Android.
- **Brave Talk** (private video calls); Wayback Machine 404 fallback (desktop-centric).
- **Brave Origin** (2026): the paid, stripped-down sibling — no Rewards/Wallet/Leo/News/Talk/sponsored images/telemetry.

3. UI/UX structure & feature placement

- **Address bar:** top by default; bottom option via Appearance settings / long-press. Shields lion icon sits in the address bar (left of the URL).
- **Shields panel (tap lion):** site name + big Shields up/down toggle; counters for trackers & ads blocked (tappable list) and blocked scripts; "Advanced controls" expander → per-site: trackers & ads level, HTTPS upgrade, block scripts, cookie blocking level, fingerprinting blocking, **Forgetful Browsing** toggle, **Block Elements** entry; link to global Shields defaults and filter management.
- **Tab switcher:** grid of thumbnails, Chromium-style; tab-groups support; toggle between normal and Private panes.

- **Main menu** (:, customizable/reorderable since v1.84) — representative default order: New tab · New private tab · History · Delete browsing data · Downloads · Bookmarks · Recent tabs · Share · Find in page · Translate · Add to Home screen · Desktop site toggle · Night Mode · Brave Leo · Brave Wallet · Brave Rewards · Brave VPN · Playlist · Print · **Customize menu** · Settings · Exit. (Feature rows like Wallet/Rewards/VPN/Leo are the ones users most often hide.)
- **Settings top-level structure (approximate order):** a **Features** block — Brave Shields & privacy · Brave Rewards · Brave News · Brave Wallet · Brave Leo · Brave VPN · Brave Playlist — followed by **General**: Search engines · Brave Sync · Passwords · Payment methods · Addresses · Privacy and security · Notifications · Site settings · Media · Downloads · Home page · Tabs · **Display/Appearance**: Appearance (incl. Night Mode, bottom bar) · New Tab Page · Theme · Accessibility · Languages · **Advanced**: About Brave.
- **Long-press / context menus**: links (open in new/private/group tab, copy, download, share, "summarize with Leo" entries in 2025+ builds); images (download, search with Brave, share); text selection (copy/share/translate/Ask Leo).
- **Gestures**: toolbar swipe = tab switch; swipe-up on bottom bar = tab grid; pull-to-refresh; swipe-to-dismiss tabs in switcher.

4. Standout features (what Brave does best)

- **Best-in-class native ad/content blocking on Android** — Rust engine + EasyList/uBO lists + scriptlets + cosmetic filtering + custom lists/rules, immune to MV3; the benchmark all other mobile browsers get measured against.
- **Block Elements tap-to-zap** (first major mobile browser with interactive element hiding).
- **Fingerprint randomization (farbling)** — genuinely differentiated anti-fingerprinting.
- **Free background video playback + built-in adblock on YouTube** — replicates YouTube Premium's two headline conveniences at \$0; the single biggest Android-specific draw.
- **Leo AI with a real privacy architecture** (proxied, no logs, no account, model choice).
- **Accountless, E2E-encrypted Sync chain**.
- **Independent search index + Ask Brave/Deep Research** — the only major browser vendor with its own index.
- **Layered navigation-tracking defenses** (De-AMP, debouncing, unlinkable bouncing, param stripping, Forgetful Browsing) that no other Android browser matches as a package.

5. Weaknesses & common criticisms

- **Crypto/Web3 clutter**: Rewards, Wallet, BAT and sponsored NTP images feel like bloat to many; so persistent that Brave productized the complaint as paid **Brave Origin** — itself criticized ("paying \$60 to remove what a group policy can disable").
- **Feature sprawl / disruptive updates**: menu and UI churn; each release adds toggles; reviews cite "inconsistency and clutter."
- **No reader mode / Speedreader on Android** (flag-only and broken) — notable gap vs Firefox/Samsung Internet.
- **No Tor tabs on Android**, despite years of requests.
- **Sync reliability**: partial history sync, occasional lost settings/passwords reports; no cloud account fallback if the chain breaks.
- **Playlist lagged years behind iOS** on Android; offline playback bugs.
- **Site breakage** under Aggressive Shields/fingerprint randomization; occasional YouTube cat-and-mouse breakage windows.
- **Past trust dings still cited**: 2020 affiliate-link autocomplete incident, opt-out sponsored images.
- **No extension support on Android** (unlike Firefox); built-ins must cover everything.

6. Recent additions (2024–2026)

- **2024**: Leo on Android (v1.63, Jan 2024); Leo model expansion; bottom-address-bar improvements.
- **Early-mid 2025**: **Rewards 3.0** redesign; **Block Elements on Android** (v1.78); custom filter lists via Android settings.
- **Mid-late 2025**: **Self-custody BAT payouts on Solana** (v1.81); **Leo quick actions/voice widget + personalization/memories/@-tab-mentions** (v1.82); VPN Smart Proxy; **Ask Brave** (Sept 2025); **customizable main menu + CSV password import + Leo PDF attachments** (v1.84).
- **Late 2025–2026**: **custom toolbar shortcuts** (v1.85); **pin website shortcuts on NTP + new Leo models** (v1.86); **YouTube improvements — better ad blocking, PiP, background play** (v1.87); dynamic Material-You theming flag; **Leo Automatic Mode**; **Brave Search Rerank** and Ask Brave **Deep Research**; **BAT Roadmap 3.0**; **Brave Origin** launched June 2026; Chromium 145→150 track.

Opera & Opera GX (Android)

1. Overview & positioning

- **Opera for Android:** full-featured Chromium/Blink browser positioned as "fast, secure & personal AI browser" — the everything-built-in alternative to Chrome: free VPN, ad blocker, AI assistant, data savings, no extensions needed. Version numbering shares the desktop release train (Opera 133 stable as of July 2026).
- **Opera GX Mobile:** "the world's first mobile browser for gamers," launched at E3 2021. Positioned on aesthetics (gaming themes, sounds, haptics), GX Corner gaming content hub, and cross-device Flow sync. Won an iF Design Award in 2026.
- Opera's Android family also includes **Opera Mini** (proxy-compression data saver, dominant in Africa/emerging markets) and formerly Opera Touch (gesture/FAB browser whose DNA lives on in GX Mobile's Fast Action Button and Flow).
- Business model: default-search deals, ad/partner content on the start page, VPN Pro subscription (~\$4/mo), and Web3/fintech (MiniPay stablecoin wallet, 10M+ wallets as of Sept 2025).
- Global mobile share is small but Opera over-indexes in Africa, Eastern Europe, and Southeast Asia on the Mini/data-saver legacy.

2. Feature inventory by category

Navigation & address bar

- Combined address/search bar at top of screen (phone layout).
- Bottom navigation bar with back/forward, Ask AI button (center, since Jan 2026), tab counter, and O-menu button.
- QR/barcode scanner built into the search field.
- Search-engine picker chip in the address bar when typing (Google, Yandex, DuckDuckGo, Amazon, etc.).
- Cryptojacking protection and fraud/malware warnings on navigation.
- Instant search suggestions and "Ask AI" entry directly from the search box (Dec 2025).

Tabs & tab management

- Three selectable tab-gallery layouts: **carousel** (classic large previews), **grid**, and **list** (most compact) — v89, May 2025, billed as "the most complete tab management system."
- **Tab Islands:** tab grouping by long-press-and-drag one tab onto another; groups can be named and managed; links opened from a page auto-cluster into an island (auto-grouping can be disabled).
- Tab search: magnifying-glass icon in tab gallery, searches page title, site name, or URL words.
- Recently closed tabs: remembers up to 100; restore individually or whole Tab Islands.
- Private tabs in a separate gallery pane; tabs synced from other devices visible via the gallery.

Home / New-tab page (speed dial, news feed)

- **Speed Dial:** grid of site shortcuts; 2026 update added customizable icon shapes (circle, square, squircle).
- **News feed** below speed dials; channel/interest picker; can be fully disabled. Contains sponsored/ad content.
- 2026 home-screen makeover: weather shortcut at top, **Google AI Mode shortcut under the search bar**, and a FIFA World Cup 2026 live-score dashboard.

Search & discovery

- Default Google with multi-engine quick switching; private search options.
- Aria/Opera browser AI answers surfaced from the search box.
- Enhanced Aria answers for recipes, shopping, and gaming categories with suggestion chips (v90).

Privacy & security (free built-in VPN)

- **Free built-in VPN:** no-log, unlimited-bandwidth, no account required; encrypts browser traffic between the device and Opera's VPN servers; hides IP/location. Coarse region choice only: **Americas / Europe / Asia**. Browser-only — does not cover other apps' traffic; technically closer to an encrypted browser proxy than a system VPN (a standard criticism).

- **VPN Pro** (subscription, ~\$4/mo): device-wide Android VPN, 3,000+ servers, 48 locations (v90, July 2025), Lightway protocol with post-quantum protection, up to 6 devices. Makes Opera "the only major browser with both free and Pro VPN."
- Cryptomining protection; pop-up blocker; cookie dialog blocker (auto-dismisses GDPR consent banners); tracker blocking bundled with the ad blocker; private browsing mode; passwords/autofill management.

Ad/content blocking (built-in)

- Native ad blocker (EasyList-based), toggle in O-menu and Settings; per-site exceptions.
- Cookie consent dialog blocker (a distinctive Opera extra).
- Blocking also accelerates pages and reduces data use.
- GX Mobile ships the same built-in ad blocker + cookie dialog blocker.

Downloads & media

- Full download manager (pause/resume, background downloads, notification controls); long-press to download media; dedicated download button in the built-in video player.
- Offline pages: save full pages for offline reading (distinct from downloads).
- Video downloads and pop-out/picture-in-picture video playback.

Reading & content

- Night mode with customizable dimming/color temperature that darkens web content (separate from UI dark theme), plus forced-dark for pages.
- Text wrap / text-size controls, forced zoom on any page.
- No full "reader mode" as prominent as Firefox's — a noted gap; Aria page summaries increasingly substitute.

Customization & appearance (themes, GX styling)

- Opera: light/dark/system app themes; accent color options; speed-dial icon shapes; **bookmark folder emojis** (v90); wallpapers, including **AI-generated wallpapers via Aria**.
- **GX Mobile**: four built-in gaming themes plus community/official **GX Mods** from store.gx.me (287+ mobile mods: themes, wallpapers, splash screens, icons, browser/keyboard sounds, background music, shaders, stickers, fonts); 2026 brought cross-device mod sync and custom shader packs.

Sync & account (Flow)

- **Flow (My Flow)**: encrypted end-to-end channel between phone and desktop Opera/GX; **no account needed** — **pair by scanning a QR code**. Share links, notes, images, videos, and files (up to 10 MB per file); acts as a self-chat/temporary shared clipboard.
- Full **Opera account sync** also available: bookmarks, open tabs, history, passwords, speed dials.

AI features (Aria)

- **Aria** → **renamed "Opera browser AI" in January 2026**. Free, **no account required**.
- Access: **Ask AI button in the middle of the bottom bar**; also from the search box.
- Capabilities: real-time web-grounded answers with sources; page-context Q&A and **page summarization**; text generation/rewriting; translation in 50+ languages; **image generation** (incl. wallpapers); **image understanding** (camera/gallery uploads); **file analysis** (PDFs/docs); voice input and spoken responses.
- Desktop-side agentic features (Browser Operator, Opera Neon at \$19.90/mo) remain desktop-first.

Gestures & shortcuts (Fast Action Button)

- **Opera GX Mobile — Fast Action Button (FAB)**: persistent thumb-reach button; press-and-slide gestures across two rows — bottom row for search / new tab / close tab / reload, top row for recent tabs — with **vibration and haptic feedback** on each element. GX's signature one-handed navigation (inherited from Opera Touch).
- Opera for Android: long-press back = history; swipe on tab gallery; swipe between speed dial and news feed; pull-to-refresh; no FAB (uses fixed bottom bar).
- Text selection quick actions (search, translate, Ask AI).

Data saving & performance

- **Data-saver heritage:** Opera Mini's server-side compression (up to 90%) built Opera's emerging-market user base; Opera for Android retains a **data savings mode** toggle that compresses pages through Opera servers.
- Ad blocking as a speed/data feature; GX desktop's RAM/CPU/network limiters do **not** exist on mobile.

Crypto/Web3 wallet

- Legacy in-browser Crypto Wallet was **phased out of Opera for Android in Dec 2024–2025**.
- Replaced by **MiniPay**: non-custodial dollar-stablecoin wallet on Celo — phone-number-based, ~1-second transfers costing ~\$0.001; 10M wallets and 270M transactions by Sept 2025. Focused on Africa/emerging markets.

Accessibility

- System font-scale respect plus in-browser forced zoom on any page and text-wrap.
- Dark/night modes; TalkBack support; Aria voice input/read-aloud doubles as an accessibility aid.
- Thinner accessibility settings than Samsung Internet.

Unique extras (GX Corner, sounds)

- **GX Corner** (GX Mobile): gaming hub — daily gaming news, release calendar, **free-game claims and deals aggregator**, personalized game recommendations (2025+).
- **Browser sounds & haptics** (GX Mobile): UI sound effects, keyboard sounds, background music mods; haptics throughout the FAB.
- Opera for Android extras: FIFA World Cup 2026 live dashboard, weather widget, emoji bookmark folders.

3. UI/UX structure & feature placement

- **Address bar + bottom bar layout:** Opera: address/search bar at **top**; persistent **bottom bar**: back, forward/search, **Ask AI (center)**, tab counter, O-menu. The true one-hand **FAB design lives in GX Mobile** — a single bottom-center button whose press-and-slide radial rows replace most chrome.
- **Tab switcher design:** Opened via tab-counter button. User-selectable **carousel / grid / list** layouts. Top-right: **tab search**. Left of "+": **recently closed** (up to 100). Separate private-tabs pane. **Tab Islands** shown as grouped clusters; long-press-drag to group. GX Mobile uses a vertical card switcher themed to the active skin.
- **Main menu (the O menu): items in order** 1. **Controls row (top):** VPN toggle · Data savings · Ad blocking · Night mode 2. **Primary items:** Settings · History · Bookmarks · Aria/Opera browser AI · Offline pages · Downloads · Flow 3. **Shortcut row:** Appearance · Sync · Passwords · Cards · Addresses 4. **Exit (power) button** at the bottom.
- **Settings: top-level sections in order (approximate):** Sign in / Opera account → **Start page** (speed dials, news feed, channels) → **Appearance** (theme, layout, address-bar options) → Search engine → Privacy & security (VPN, ad blocker, cookie dialog blocker, tracking, site settings, passwords) → **Browser** (data savings, Tab Islands auto-grouping, text wrap, downloads location) → **Notifications** → **Sync and backup** (incl. Flow/QR pairing) → About/help.
- **Long-press / context menus:** Links: open in new tab / background / private tab / copy / share / download / send to Flow. Images: download, share, copy, search-with, Ask AI. Selected text: copy, search, translate, Ask AI. Speed-dial tiles: edit, move, delete. Tab gallery: long-press-drag to build Tab Islands.
- **Gestures:** Swipe up/down between speed dial and news feed; pull-to-refresh; swipe-to-close in tab gallery; long-press back for history. GX Mobile: FAB press-and-slide gestures with haptic ticks.

4. Standout features (what Opera does best)

- **Free, unlimited, no-account VPN inside the browser** — still unique among major browsers; paired with a real device-wide VPN Pro upsell.
- **Everything built-in:** ad blocker + tracker blocker + cookie-dialog blocker + VPN + AI with zero extensions or accounts.
- **Aria / Opera browser AI free with no login** — one of the most complete free mobile AI integrations.
- **Flow:** the slickest no-account phone↔desktop pipeline (QR pairing, E2E-encrypted links/notes/files).
- **Best-in-class mobile tab management** (three layouts, Tab Islands, tab search, 100-tab undo).
- **Data-savings heritage:** compression modes unbeatable on cost-sensitive networks.
- **GX Mobile's personality:** the only mainstream mobile browser with themes+sounds+haptics+FAB one-handed control and a gaming content hub; iF Design Award 2026.

5. Weaknesses & common criticisms

- **Chinese-ownership trust deficit:** majority stake held by Kunlun Tech consortium (since 2016); recurring data-sharing concerns despite Norway HQ.
- **"VPN" is really a browser proxy:** free tier protects only Opera traffic, coarse 3-region choice; unsuitable for whole-device protection unless you pay for VPN Pro.
- **Ad-funded start page:** sponsored speed dials, news-feed ads, and partner shortcuts read as bloat; requires manual decluttering.
- **Feature bloat / upsell fatigue:** AI, crypto/MiniPay, World Cup dashboards, VPN Pro promos — "kitchen-sink" criticism; GX's novelty features seen as gimmicks by non-gamers.
- **Slower security patch cadence than Chrome.**
- **No mobile extensions;** no true reader mode; weaker accessibility settings.
- GX Mobile lacks desktop GX's actual utility features (RAM/CPU/network limiters) — largely cosmetic differentiation.
- Data-savings compression routes traffic through Opera servers — a privacy trade-off.

6. Recent additions (2024-2026)

- **Dec 2024:** Opera GX "facelift"; Crypto Wallet phase-out on Android begins.
- **Apr–May 2025 (v89):** complete Tab Management overhaul — carousel/grid/list layouts, Tab Islands with auto-grouping, tab search, 100 recently-closed restore.
- **Jul 2025 (v90):** VPN Pro revamp — 48 locations, Lightway protocol with post-quantum protection; enhanced Aria category answers; emoji bookmark folders.
- **May 2025 (GX):** GX Corner upgrade — personalized game recommendations, real-time Steam/Epic deals.
- **Dec 2025:** Ask AI accessible directly from the Android search box; rebuilt accountless "Opera AI."
- **Jan 2026: Aria renamed "Opera browser AI"** — Ask AI button centered in the bottom bar; page summaries, file/PDF analysis, image understanding, voice, AI wallpaper generation.
- **Jun 2026 (GX "Summer" update):** 2x faster startup, refreshed shader/sidebar panels, mini player, new ad blocker, smarter AI; iF Design Awards.
- **Jun–Jul 2026:** Opera Android home-screen makeover — weather shortcut, speed-dial icon shapes, **Google AI Mode shortcut under the search bar**, FIFA World Cup 2026 live-score dashboard.

Niche power browsers: Vivaldi · DuckDuckGo · Kiwi (Android)

Vivaldi (Android)

1. Overview & positioning

Vivaldi on Android is the mobile companion to Vivaldi's desktop browser, built on Chromium/Blink by Vivaldi Technologies (Norway/Iceland) and positioned as the most feature-rich, customizable power-user browser on Android — "everything is an option." Actively developed as of mid-2026 (v8.0 released May 28, 2026). Free, no account required, funded by search/bookmark partner deals rather than user data.

2. Feature inventory by category

- **Navigation & address bar:** Combined search/address field; multiple built-in search engines with "search engine nicknames"; **custom search engines** (7.7, Nov 2025 — long-press a site's search field > "Add as Search Engine"); search-with-selected-text; address bar movable top or bottom; pull-to-refresh can be disabled (7.8).
- **Tabs & tab management:** Desktop-style **visible Tab Bar on phones and tablets** (optional); **Tab Stacks** (drag tab onto tab to group); **Two-Level Tab Bar** — world-first two rows of mobile tabs (Vivaldi 5.0): stacked tabs shown on a second row; renamable tab stacks; choice of **Accordion Tabs vs Two-Level** stack display (Accordion arrived on Android in 8.0); **pinned tabs on Android** (7.8); Tab Switcher with sections for normal/private/synced/recently-closed tabs; tab search (7.1); close-tab-by-swipe; clone tab, background tab opening.
- **Home / New-tab page:** **Speed Dial** start page with folders/groups, customizable columns and layout; unified Start Page combining **Widgets and Speed Dials** (7.7); "Add Page to..." menu (Speed Dial / bookmark / reading list / home screen).
- **Privacy & security:** Built-in **tracker and ad blocker** (per-site configurable, blocking levels: off / trackers / trackers+ads, custom filter lists); **Privacy Dashboard** showing blocked ads/trackers counts (7.8); private tabs; option to use third-party autofill/password manager (7.8); no user profiling by vendor; end-to-end encrypted sync; cookie-consent-blocker ("Cookie Crumbler") lists.
- **Ad/content blocking:** Integrated blocker (EasyList/EasyPrivacy-style lists plus custom list import) — no extensions needed; per-site exceptions.
- **Downloads & media:** Built-in download manager accessible from Panels; **integrated PDF viewer** (8.0 — PDFs open in-browser).
- **Reading & content:** **Reader View** (redesigned in 7.1); **Reading List; Notes** — full note manager with titles, synced with desktop notes; **Vivaldi Translate** — built-in private translator (Lingvanex-powered, hosted on Vivaldi servers); **page capture / full-page screenshot** from the menu.
- **Customization & appearance:** **Address Bar + Tab Bar position: top or bottom** (independently configurable); customizable **toolbar shortcut button**; themes, accent colors, app icon options; per-site UI theming from page colors; hide navigation bar on scroll; improved forced dark mode for web pages (7.7); disable Tab Bar entirely.
- **Sync & account:** **Vivaldi Sync** — end-to-end encrypted with a separate encryption password, free Vivaldi.net account: bookmarks, Speed Dials, passwords, autofill, history, tabs, notes, reading list, search engines.
- **AI features:** Deliberately none — Vivaldi has publicly taken an anti-"AI-in-browser" stance, marketed as a feature ("browse without AI interference").
- **Gestures & shortcuts:** Drag-down from top toolbar gestures; swipe-to-close tabs; swipe between tabs on address bar; long-press context actions throughout.
- **Extensions: Not supported** on Android (a top user complaint) — Vivaldi argues the built-in feature set replaces most extensions.
- **Unique extras:** Notes with sync; full-page capture; built-in translate; Reading List; Panels sidebar (Bookmarks, History, Notes, Downloads); granular per-site performance controls (7.7); Chromecast support.

3. UI/UX structure & feature placement

- Address bar top **or** bottom; optional desktop-style Tab Bar (top), which can also move with the address bar; on tablets/landscape, bottom-toolbar icons merge into the top toolbar for a desktop-like layout.
- Bottom toolbar (phones): Panels button (left) opens slide-out panel for Bookmarks/History/Notes/Downloads; back/forward, tab-switcher button, customizable shortcut button, and the **"V" Vivaldi menu** (a long scrollable menu of actions).
- Distinctive UX: the only major Android browser that reproduces a desktop tab strip + tab stacks on a phone; Speed Dial-centric start page; settings depth is unmatched (and notoriously deep).

4. Standout features

- Two-Level Tab Bar / Tab Stacks with Accordion mode — genuinely unique on mobile.
- Full toolbar/address-bar position customization predating Chrome's own bottom-bar experiments.
- Built-in ad/tracker blocker + Privacy Dashboard, notes sync, translate, page capture — "no extensions needed" bundle.
- E2E-encrypted sync without a Big Tech account.
- Explicitly AI-free positioning (differentiator in 2026).

5. Weaknesses & criticisms

- No extension support on Android; mobile lags desktop on advanced tab management/workspaces.
 - Complexity/discoverability: features buried behind multiple menus/submenus; overwhelming settings for casual users.
 - Performance: heavier than stock Chromium builds; sluggish on low-end devices.
 - Smaller ecosystem/market share means occasional site-compat quirks; dated visual design per some reviewers.
-

DuckDuckGo Browser (Android)

1. Overview & positioning

DuckDuckGo Private Browser is a privacy-first Android browser built on the system **WebView** (Blink) around DuckDuckGo's search engine, positioned as "privacy, simplified" — protection by default rather than power-user configuration. Free, with an optional paid Privacy Pro subscription (VPN, advanced Duck.ai models, identity protection). Deliberately minimalist. **Notable for Andromeda: it proves a serious browser can ship on system WebView.**

2. Feature inventory by category

- **Navigation & address bar:** Single combined search/address bar wired to DuckDuckGo Private Search; address bar position **top or bottom**; anonymous local autocomplete.
- **Tabs & tab management:** Basic tab switcher (grid/list); **swipeable tabs** (swipe on address bar to switch); tab previews; no tab groups — intentionally simple.
- **Home / New-tab page:** Minimal: search box + **Favorites** grid; recently a cleaner "focused" layout. No widgets/speed-dial folders.
- **Privacy & security** — the core of the product:
- **3rd-Party Tracker Loading Protection** (blocks tracking scripts before load)
- **App Tracking Protection** (Android-exclusive: a local on-device VPN service that blocks third-party trackers inside *other apps* — no traffic leaves the device)
- **Fire Button** (one-tap flame icon burns all tabs + browsing data, with animation; per-site fireproofing to keep logins)
- **Cookie Pop-Up Management** (auto-answers consent banners), **Smarter Encryption** (HTTPS upgrading), **Global Privacy Control (GPC)**, **Email Protection** (@duck.com aliasing), **Scam Blocker** (phishing/malware/scam-site blocking, expanded 2025), link tracking-parameter stripping, referrer trimming, fingerprinting mitigations, embedded-content protection.
- **Ad/content blocking:** Historically no general ad blocker (ads only disappeared as a side effect of tracker blocking). As of **July 2026**: built-in **video ad blocking incl. YouTube**, based on uBlock Origin community filter lists — manual enable on Android (Settings > Ad Blocking); DDG signaled expansion toward broader ad blocking.
- **Downloads & media:** Basic download manager; **Duck Player** — built-in YouTube player enforcing YouTube's strictest embedded privacy settings (no targeted ads, views don't feed recommendations).
- **Reading & content:** No reader mode, no reading list, no translate — notable gaps.
- **Customization & appearance:** Light/dark/system theme; address bar position; show/hide Duck.ai buttons; app icon variants; otherwise minimal by design.
- **Sync & account:** **Sync & Backup** — end-to-end encrypted, **no account needed** (device-pairing via QR code + recovery PDF); syncs bookmarks, passwords (built-in password manager), and Email Protection settings.
- **AI features:** **Duck.ai** — private access to multiple chat models (free tier: Claude Haiku, GPT minis, Mistral, open models; Privacy Pro tiers add frontier models) with anonymized proxying, no training on chats, chats stored locally only; optional AI-assisted search answers; **fully optional** — a master toggle hides all AI UI.

- **Gestures & shortcuts:** Swipe address bar to switch tabs; pull-to-refresh; long-press for link/image actions; Fire Button as the signature one-tap action.
- **Extensions:** None.
- **Unique extras:** App Tracking Protection (unique on Android); GPC; @duck.com email aliases; per-site Privacy Dashboard (tap the shield); paid **Privacy Pro:** VPN, Personal Information Removal, Identity Theft Restoration.

3. UI/UX structure & feature placement

- Single bar (top by default, bottom optional) + three-dot overflow menu; shield icon in the bar opens the per-site privacy dashboard (tracker list, protections toggle); **flame/Fire Button** prominent in the toolbar/menu — burning data triggers the signature fire animation.
- Tab icon opens a simple switcher; Settings organized around protections. App Tracking Protection runs as a local VPN with a notification of blocked in-app trackers.
- Distinctive UX: radical simplicity — closer to a search app than a power browser; zero-config privacy with the Fire Button ritual as its most memorable interaction.

4. Standout features

- App Tracking Protection (system-wide, on-device in-app tracker blocking) — unique on Android.
- Fire Button + fireproofing; per-site privacy dashboard.
- Duck.ai multi-model private AI chat with a true global off switch — the most opt-out-friendly AI integration of any major browser.
- Duck Player + new uBlock-list-based YouTube ad blocking (2026).
- Accountless E2E-encrypted sync; Email Protection aliases; GPC pioneer.

5. Weaknesses & criticisms

- **2022 Microsoft tracker scandal:** browser initially exempted Microsoft trackers due to the Bing syndication deal — a lasting trust dent.
- Feature-poor as a *browser*: no extensions, no reader mode, no translate, no tab groups; power users outgrow it quickly.
- Fingerprinting protection is API-spoofing rather than robust blocking; researchers showed persistent localStorage surviving data clearing on Android.
- App Tracking Protection occupies Android's one VPN slot (can't run alongside a real VPN) and can break some apps.
- YouTube ad blocking still not default-on for Android as of July 2026.

Kiwi Browser (Android) — legacy

1. Overview & positioning

Kiwi Browser was a Chromium-based Android browser created in 2018 by solo developer Arnaud Granal, famous as the **first mobile browser with full desktop Chrome-extension support**. **Discontinued early 2025** (developer burnout; ~500k active users); pulled from Google Play, GitHub repo archived, extension-support code **donated to Microsoft Edge for Android**. Covered as legacy.

2. Feature inventory (legacy, final builds)

- **Navigation:** Chrome-style omnibox; revived "Chrome Home" **bottom address bar** option; custom search engines. (Caveat: search-redirect controversy, \$5.)
- **Tabs:** Standard Chrome tab switcher/grid, tab groups inherited from Chromium.
- **Home / NTP:** Chrome-like NTP with configurable background, option to remove sponsored/suggested content.
- **Privacy & security:** Pop-up blocker, **cryptojacking/cryptominer blocking**, blocking of intrusive notification prompts, ability to **disable AMP**; otherwise inherited Chromium's model. Not a privacy-hardened fork.
- **Ad/content blocking:** Built-in ad blocker (plus most users layered uBlock Origin as an extension — its real superpower).
- **Downloads & media:** Chromium download manager; **background video/audio playback** (YouTube audio with screen off — a headline feature).

- **Reading & content:** Chromium simplified/reader view; built-in Translate; night/dark rendering.
- **Customization: Advanced Night Mode** — customizable forced dark mode with contrast/grayscale controls (AMOLED-friendly), one of the earliest and best implementations; bottom toolbar option; export/import settings.
- **Sync: None** — no cross-device sync of its own; bookmarks import/export only. A frequently cited gap.
- **Extensions — the signature:** Installed extensions **directly from the Chrome Web Store** with a near-desktop `chrome://extensions` page; supported uBlock Origin, Dark Reader, Tampermonkey, Bitwarden, Stylus, SponsorBlock, etc.; most reliable with **Manifest V2** extensions — which Google is phasing out, so the frozen final build ages badly.
- **Unique extras:** Open-sourced (2023); served as the reference implementation that seeded extension support in **Microsoft Edge for Android**.

3. UI/UX structure

- Essentially **Chrome's UI with power toggles**: top omnibox with optional bottom bar; standard three-dot menu extended with extension entries, night mode, translate; standard Chrome tab grid.
- Extensions surfaced via the overflow menu and a desktop-style `chrome://extensions` page.
- Distinctive UX: zero learning curve for Chrome users — its identity was *removing Chrome's mobile limitations* (extensions, background play, night mode, bottom bar).

4. Standout features

- Full Chrome Web Store extension support on a phone — uBlock Origin + Tampermonkey on mobile was its killer app.
- Background YouTube playback for free; advanced night mode; cryptominer blocking; bottom address bar years before Chrome.
- Legacy impact: its extension code lives on in Microsoft Edge for Android.

5. Weaknesses & criticisms

- **Discontinued and frozen:** no security patches on an aging Chromium base — actively unsafe to keep using in 2026.
- **Monetization/privacy controversy:** documented claims that Kiwi shipped fake "Yahoo/Bing" search engine entries that routed queries through Kiwi's own servers for revenue.
- Chronically behind Chromium upstream even while alive; solo-maintainer bus factor was the ultimate cause of death.
- No sync; MV2 dependence means even nostalgia use degrades.
- Successor landscape is fragmented: Edge for Android (curated extensions), Quetta, Lemur/Mises forks, or Firefox for Android — none replicates Kiwi's open model.

AI-first browsers (2025-2026 wave)

1. The landscape

Who the players are (as of mid-2026):

Player	Product(s)	Status mid-2026
Perplexity	Comet (desktop + iOS + Android)	Category leader on mobile; went fully free March 18, 2026 on all platforms; raised \$200M for Comet in June 2026
The Browser Company / Atlassian	Dia (macOS) + Arc Search (iOS/Android legacy)	Acquired by Atlassian for \$610M cash (closed Oct 2025); Arc sunset; Dia mobile not yet shipped — "Arc Search-inspired mobile updates" promised for 2026
OpenAI	ChatGPT Atlas (macOS)	Launched Oct 2025, discontinued — sunsets Aug 9, 2026 ; agentic browsing folded into the ChatGPT app instead. Mobile version never shipped
Opera	Opera AI (formerly Aria) ; Opera Neon (premium agentic, \$19.90/mo)	Aria rebranded "Opera AI" late 2025, free; Neon publicly available Dec 2025
Google	Gemini in Chrome (desktop + iOS/Android)	Sidebar + agentic "auto-browse" rolled out Jan 2026; agentic features gated behind AI Pro/Ultra
Samsung	Samsung Browser "Browsing Assist"	Rebuilt browser launched March 26, 2026; conversational + agentic AI on Android and Windows, powered by Perplexity APIs ; ~1B device reach; US/Korea only for agentic features
Microsoft	Edge Copilot Mode	Strong for M365 users, less compelling outside that ecosystem
Brave	Leo	Privacy-first assistant (anonymized proxy, local model option); deliberately non-agentic
Apple	Safari	AI overhaul reportedly paused ; Apple pivoting to Gemini-powered Siri instead of an AI Safari
Smaller entrants	SigmaOS , Genspark (169+ on-device open-weight models), Fellou (agentic, editable workflow plans)	Niche; none has meaningful mobile presence

What "AI browser" means in 2026: two tiers — (1) **assistant browsers** (chat sidebar, summarize, ask-the-page: Leo, Edge Copilot, Opera AI, Gemini-in-Chrome baseline) and (2) **agentic browsers** (AI navigates, fills forms, completes multi-step tasks: Comet, Dia, Neon, Chrome auto-browse). The 2026 defining features are cross-tab context, browser "memories," and agent mode.

Adoption reality vs hype: - Chrome still holds ~65–71% global share. Dedicated AI browsers remain **niche**; the real shift is incumbents (Chrome, Edge, Samsung) absorbing AI features, which happened fast in H1 2026. - OpenAI killing Atlas after ~9 months is the wave's biggest deflation signal: the standalone "AI browser" may be a feature, not a product. - Counter-signal: Perplexity making Comet free everywhere + the Samsung distribution deal (1B devices) shows the mobile battle is fought on **distribution and zero price**, not subscriptions. - Barriers: learning curve of new UI paradigms, battery/compute cost on low-end phones, and unsolved prompt-injection security. - Business models diverging: free + publisher revenue share (Comet Plus pays **80% of revenue to publishers**) vs. \$19.90/mo premium (Neon) vs. enterprise SaaS (Atlassian/Dia).

2. Per-browser breakdown

Perplexity Comet (mobile — the reference implementation)

- **Core AI:** Assistant that (a) understands the current page ("ask-the-page"), (b) runs **multi-step agentic tasks across tabs**, (c) exposes Perplexity's citation-backed answer engine as the default search. **Cross-tab summarization.** Voice mode. Deep Research in-browser. On iOS, agent tasks run in a **cloud virtual browser** — cookies temporarily transferred, screenshots stream back.
- **Non-AI:** Built-in ad/tracker blocker; Chromium-based on Android/desktop with Chrome extension support (desktop only — no extensions on mobile); reviewers found it renders pages faster than Chrome.
- **UI/UX:** "One-Thumb" philosophy — omnibox and **Assistant Drawer at the bottom of the screen**; the drawer is a dynamic overlay summoned with one tap so you keep visual context of the page. Bottom address bar that minimizes on scroll, **prominent Assistant button center of the address bar**, voice button on its right, **long-press a link → "Ask Comet"**, a Library screen for past AI conversations. During agent runs, **blue screen edges signal "AI is in control"** and a tappable step-by-step action log provides transparency/intervention.
- **Pricing:** Free (since March 2026). Optional **Comet Plus \$5/mo** (premium publisher content).
- **Platforms:** macOS, Windows, **Android (Nov 2025)**, iOS/iPadOS.

Arc Search / Dia (The Browser Company, now Atlassian)

- **Arc Search (iOS + Android, the 2024-25 mobile pioneer, still live):**
- Core AI: **"Browse for Me"** — AI builds a custom results webpage per query (facts, images, citations); **pinch-to-summarize** gesture (two-finger pinch folds the page origami-style with haptics while a summary generates); voice search; "Call Arc" (search by talking as if on a phone call); toggle to flip between Google and Browse-for-Me.
- Non-AI: rebuilt ad/tracker blocking (2x effectiveness), Google Translate integration, home-screen widget, Arc Sync.
- Pricing: free.
- **Dia (the strategic future):**
- macOS-only, public since Oct 2025. Chat-with-your-tabs, **"Skills"** (user-created custom AI automations), personal "work memory," planned deep Jira/Linear/Confluence context under Atlassian.
- **No mobile app as of mid-2026;** "Arc Search-inspired mobile updates come to Dia during 2026."
- Pricing: free tier with limits; Pro subscription for unlimited AI use.

Opera AI + Opera Neon

- **Opera AI (ex-Aria), free, embedded everywhere incl. Android:** tab-aware chat, page/article summarization, YouTube video summarization + translation, PDF summarization, free image generation, voice input, real-time web access. Late-2025 upgrade: faster engine, local model options.
- **Opera Neon (separate premium product, desktop): \$19.90/mo** — agents "Chat," **"Do"** (task completion), and **"Make"** (builds mini web apps/documents autonomously). Reception: widely judged overpriced vs free Chrome/Comet.

ChatGPT Atlas (OpenAI) — cautionary tale

- Chromium macOS browser (Oct 2025): ChatGPT sidebar everywhere, **browser memories**, tab-level conversations, paid-only **Agent Mode**. Windows/iOS/Android promised, never shipped. **Killed July 2026 (service stops Aug 9, 2026)** — agentic browsing merged into the ChatGPT desktop app. Data point: even OpenAI couldn't justify a standalone browser.

Others (brief)

- **Samsung Browser Browsing Assist:** summarize, translate, read-aloud, natural-language **tab management and history navigation**, agentic actions; Perplexity-powered; free; the biggest OEM-default AI browser play.
- **Gemini in Chrome (mobile):** address-bar Gemini icon for ask-the-page/summarize; desktop gets sidebar + agentic auto-browse (find item, apply coupon, buy) for AI Pro/Ultra subscribers.
- **Brave Leo:** summarize, ask-the-page via privacy proxy, optional local models; explicitly no agentic actions.
- **Fellou:** agentic with **user-editable workflow plans before execution** — notable UX idea; also had prompt-injection vulnerabilities.

3. Common AI feature patterns

Every AI browser converges on this stack (roughly in order of ubiquity):

1. **Summarize** — current page, then video (YouTube), PDFs, and (differentiator tier) **all open tabs at once**.
2. **Ask-the-page / contextual chat** — Q&A grounded in current tab content; the table-stakes feature.
3. **AI omnibox / answer-engine search** — address bar accepts natural-language questions and returns a generated, cited answer.
4. **Agentic actions / "browse for me" tasks** — multi-step navigation, form filling, shopping, booking. Almost always with a visible step log and interrupt control; on iOS, executed via cloud virtual browsers.
5. **Cross-tab context + tab management by language** — "compare my open tabs," "close everything about X."
6. **Memory / personalization** — browser remembers your history/context for the assistant.
7. **Voice mode** — realtime speech assistant.
8. **Writing help in text fields** and generation extras — the least essential tier.
9. **Non-AI accompaniments that ship with nearly all of them:** built-in ad/tracker blocking, cross-device sync, bottom-bar mobile ergonomics.

4. UI/UX patterns for AI in browsers

- **Bottom address bar as AI host (mobile standard):** Comet puts the Assistant button dead-center in a bottom omnibox ("one-thumb reach"); the bar shrinks on scroll.

- **Assistant drawer / bottom-sheet overlay:** the dominant mobile pattern — AI slides up over the page so the page stays visible; not a separate screen.
- **Sidebar panel (desktop pattern):** largely irrelevant on phones but shapes tablet layouts.
- **Gestures:** Arc Search's **two-finger pinch-to-summarize** (origami fold animation + haptics) is the canonical gesture experiment — praised for delight, criticized for summary quality.
- **Long-press context menus:** long-press a link → "Ask Comet"; text-selection → AI actions (summarize/translate/explain) is common.
- **AI-generated results as a page:** Browse for Me renders the answer *as the destination webpage* with a toggle back to Google.
- **Agent-mode takeover signals: colored screen border (Comet's blue edges) while AI controls the browser,** plus a reviewable step-by-step action log and a stop/intervene affordance; Fellou goes further with an editable plan preview *before* execution. This transparency layer is now considered mandatory UX for agentic modes.
- **Dedicated screens:** conversation Library/history; voice mode as a full-screen call-style UI.
- **Address-bar icon entry point:** Gemini in Chrome iOS uses a small icon in the address bar — the lowest-friction incumbent pattern.

5. What actually works vs gimmicks (per reviews/user sentiment)

Works (consistent praise): - **Summarization + ask-the-page** — the daily-driver features; reviewers who kept an AI browser kept it for research/reading workflows. - **Cited answer-engine search as default** — genuinely replaces some Google queries. - **Cross-tab context** ("compare these tabs") — repeatedly called the surprise killer feature. - **Speed:** Comet rendered common sites faster than Chrome in independent testing — AI browsers are not inherently slow. - **Voice mode on mobile** for hands-free lookup. - Reviewer verdicts on Comet: "faster than expected on simple tasks, more useful than expected on research, **rougher than expected on agentic flows.**"

Gimmicks / broken (consistent criticism): - **Agentic task completion** — the marquee feature is the weakest: slow, fragile on real-world sites; reviewers advise supervising every run. - **Pinch-to-summarize** — beloved animation, mediocre summaries. - **Image/video generation inside a browser** — widely tagged as padding. - **Security is the category's open wound:** Brave's research showed **indirect prompt injection is systemic** — Comet executed hidden instructions from page content and even from **invisible text in screenshots (OCR steganography attack, Oct 2025)**. "No major AI browser has fully solved prompt injection as of 2026." Practical user posture: use AI for reading/research, keep agents away from logged-in/financial contexts.

6. Requirements to build these (APIs, on-device models, costs) — brief and practical

Rendering layer: Android → WebView or a Chromium fork (WebView is sufficient for assistant-tier features).

AI layer options: 1. **Cloud LLM APIs** (what Comet/Opera/Samsung use): any frontier or mid-tier model API for summarize/ask-page; Perplexity's **Sonar API** is literally what powers Samsung's Browsing Assist. Typical costs: **\$0.01–0.10 per page interaction**; agentic multi-step tasks \$0.10–0.30 per task on frontier models. Cheap models (Gemini Flash-class, Haiku-class) cut summarization to fractions of a cent per page. 2. **On-device (free inference, private):** Android **ML Kit GenAI APIs on AI Core/Gemini Nano** — Summarization API (up to ~3,000 words, EN/JA/KO), Proofreading, Rewriting, Image Description, and the newer **Prompt API** for custom prompts. Zero cloud cost; runs on Pixel 9/10, Galaxy S25, and other flagships — **flagship-only coverage**, so a cloud fallback is still required. 3. **Hybrid** is the practical 2026 pattern: on-device for summarize/translate, cloud for search-answers and agents.

Agentic stack (if attempted): page → structured text extraction (Readability-style), an orchestration loop, action transparency UI (step log, colored border, stop button) and hard guardrails — treat all page content as untrusted input (prompt injection), never let the agent act on sensitive origins without confirmation.

Build cost reality: assistant-tier features (summarize + ask-page + AI omnibox via one API) are a small team's work; full agentic browsers are \$80k–\$1.5M builds commercially. Non-AI table stakes to be credible: ad blocking, bottom-bar UX, sync.

Mobile browser UI/UX structure patterns

Research compiled July 2026 from vendor blogs, UX teardowns, tech press, and Mozilla/Google/Samsung support documentation. Focus: Chrome, Safari iOS, Firefox, Edge, Samsung Internet, Brave, Opera, Arc.

1. Address bar placement wars

The reachability argument. The driver of the entire debate is thumb-zone research (originating with Steven Hooper's 2013 grip studies): ~49% of users hold phones one-handed, ~75% of interactions are thumb-driven, and on 6.5"+ displays the top edge is the hardest-to-reach "red zone" while the bottom third is the natural "green zone." Every vendor cites one-handed use as the rationale for bottom placement.

Safari's bottom-bar story (the cautionary tale + the template). Apple moved Safari's URL bar to the bottom in iOS 15 (2021) as a *floating* bar atop page content. Backlash was severe: the floating bar occluded page elements, and often-used actions were buried in a three-dot menu. By beta 6 Apple docked the bar *below* content and added a top-placement toggle. Crucially, once optional-and-docked, bottom placement stuck and became the iOS default. In iOS 26's Liquid Glass redesign Safari offers **three layouts — Compact (default, minimal bottom pill that shrinks on scroll), Bottom (full-width bar + toolbar), Top** — with edge-to-edge page content. Lessons encoded: bottom wins for reach, docked beats floating, and users demand a choice.

Chrome's bottom omnibox rollout. Google prototyped a bottom bar ("Chrome Home") in 2016 and cancelled it. Chrome iOS got an optional bottom address bar in 2023. On Android: testing began Oct 2024, shipped Chrome 134/135 (Mar–Apr 2025), wide rollout complete by ~Chrome 138 in July 2025. Discovery mechanism: **long-press the omnibox → "Move address bar to bottom."** Top remains Chrome's default; bottom is opt-in.

Everyone else. - **Firefox Android:** top/bottom toggle since the 2020 Fenix rewrite; bottom is the modern default for new installs. - **Edge Android:** optional bottom address bar (shipped Feb 2025). - **Samsung Internet:** historically the split model — address bar top, action toolbar bottom — now offers full bottom placement. - **Vivaldi Android:** fully configurable — address bar *and* tab bar each top or bottom. - **Brave Android:** bottom toolbar option; criticized because moving the toolbar bottom historically left the address bar at top (split). - **Opera:** bottom navigation and the one-handed Fast Action Button were the founding premise of Opera Touch/GX. - **Arc Search:** all controls on the bottom edge.

Industry consensus trend (mid-2026): configurability is table stakes; **bottom is the ergonomic consensus and the default for newer/redesigned browsers**, while incumbents with muscle-memory legacies default top but ship bottom as an option. The old split model (address top / tools bottom) is dying in favor of a unified bottom cluster.

2. Toolbar anatomy

Persistent-button counts converge on 3–5 plus the address field.

Browser	Persistent toolbar contents	Customizable?
Chrome Android	Omnibox + tab-switcher square + 3-dot menu; optional one configurable "toolbar shortcut" slot	One slot
Safari iOS	Bottom bar: back, forward, share, bookmarks, tabs around/below URL; iOS 26 Compact collapses these behind a pill menu	Layout choice, not per-button
Firefox Android	Address bar + tabs + menu; 2024–25 update added optional back/forward navigation bar	Position + layout + shortcut
Edge Android	Bottom bar ~5: home, tabs, ellipsis menu center , share	Limited; A/B-tested
Samsung Internet	Bottom bar: back, forward, home, bookmarks, tabs, menu (up to 7)	Fully reorderable drag-and-drop — the gold standard
Brave Android	Optional bottom toolbar; Shields button in the address bar	Toolbar on/off
Vivaldi Android	Back, forward, home, tabs, menu + one swappable shortcut slot	One slot + bar positions
Opera Android	Bottom bar: back, search, Fast Action Button (center, haptic) , tabs, menu	FAB vs standard choice

Patterns: (a) tabs-button + menu-button are universal and always adjacent to the address bar; (b) back/forward as persistent buttons is optional on Android (system back gesture) — yet Firefox and Samsung ship them anyway for discoverability; (c) the emerging compromise on customization is **one user-configurable slot** (Chrome, Vivaldi, Firefox) rather than Samsung-style full reordering; (d) center position is used for the highest-value action (Edge's menu, Opera's FAB).

3. Tab switcher patterns

- **Grid of cards is the dominant pattern.** Chrome uses a 2-column grid (since 2021); Safari a vertically scrolling grid; Firefox offers **grid or list**; Samsung Internet uniquely offers **three views: List, Stack, or Grid**. Arc Search is the outlier: a horizontal deck identical to the iOS app switcher.
- **Tab groups UX:** Chrome shows groups as stacked cards with a tab count; while browsing a grouped tab, a bottom strip gives quick access to siblings. Safari treats groups as **workspaces** — you inhabit one group at a time. Firefox shipped color-coded tab groups to Android in 2025 with drag-to-group. Opera's equivalent is auto "Tab Islands."
- **Private-tab separation:** universal pattern is a **segmented toggle/pane inside the switcher** plus a **full theme shift to dark/branded-purple** to signal mode — Firefox made private windows dark-by-default explicitly "to increase the feeling of privacy and easily distinguish" modes.
- **Tab count badges:** the toolbar tabs button is a rounded square with the live count. Chrome caps it at two characters — at 100+ it becomes **"D"** (smiley), a deliberate constraint-driven easter egg.
- Swipe-to-close on cards is standard everywhere.

4. Menu architecture

Three competing architectures, now converging:

1. **Long vertical list (Chrome, Brave, Vivaldi).** Chrome's 3-dot menu: a top **icon action row** (forward, bookmark, download, refresh in circular buttons — Chrome 150 added back) followed by a scrolling list ordered: New tab / Incognito → History, Delete browsing data, Downloads, Bookmarks, Recent tabs → Share, Find in page, Translate, Add to Home screen, Desktop site → Settings, Help. Ordering logic: creation actions first, library second, page actions third, app-level last.
2. **Icon grid bottom sheet (Samsung Internet, Edge, Opera).** Edge's menu is a grid of icon shortcuts, expandable and **user-rearrangeable**; Samsung's grid is fully customizable via drag-and-drop. Grids scale to many features but hurt scannability for infrequent users.
3. **Redesigned category menu (Firefox 141, 2025).** Back/Forward moved *into* the menu header; a single icon row for the frequent four (History, Bookmarks, Downloads, Passwords); collapsible Extensions section; remaining items grouped by category. Reception mixed — praised for fewer taps, criticized for screen space.

Menu vs settings vs long-press division of labor (consistent across all eight): the **menu** holds session/page actions used weekly (new tab, history, share, find, desktop site); **settings** holds persistent preferences visited rarely; **long-press** holds contextual actions — links, images, address bar, NTP shortcuts. Emerging hybrid consensus: **icon action row for high-frequency verbs + short categorized list below** (Chrome M3 and Firefox 141 arrived at the same shape independently).

5. New-tab/home page patterns

- **Chrome:** search-first — logo + omnibox + editable shortcut tiles + **Discover feed** below the fold. The feed is the single biggest complaint; removable via toggle.
- **Edge:** the feed-first MSN NTP is the industry's most complained-about NTP. In 2026 Microsoft is replacing it with the **Copilot new tab page** (prompt box + quick links + optional feed).
- **Firefox:** shortcuts (incl. sponsored) + "Stories." Everything toggleable; long history of complaints about defaults.
- **Samsung Internet:** **Quick access** editable shortcut grid — essentially speed dial; regional news widgets optional.
- **Brave:** full-bleed background image (incl. sponsored), **privacy stats block** (ads/trackers blocked, bandwidth saved, time saved) as brand reinforcement, top sites, optional Brave News.
- **Opera:** the original **speed dial** grid plus optional news feed.
- **Arc Search:** no home page at all — opens directly into a search field with keyboard raised; speed-to-input as the entire NTP philosophy.
- **Sentiment pattern:** feeds monetize but generate the loudest negative sentiment; shortcut grids + search box are near-universally accepted; **user-editable = accepted, algorithm-pushed = resented**. Sponsored tiles/backgrounds draw less anger than sponsored content feeds.

6. Gesture vocabularies

The standard cross-browser set (safe to assume users know): - **Pull-to-refresh** — universal on Android. - **Horizontal swipe on the address bar to switch tabs** — Chrome, Safari, Samsung, Brave, Firefox, Edge; near-universal. - **Edge swipes for back/forward history** — Android predictive-back gesture (browsers must not fight it). - **Swipe a card sideways to close** in the tab switcher — universal. - **Long-press** for all contextual menus.

Safari extras: swipe *up* on the address bar → tab overview; pinch anywhere → tab overview.

Novel/signature gestures: Chrome's long-press-omnibox → move bar to bottom (gesture as settings discovery); Opera's Fast Action Button press-and-slide with haptics; Arc Search's swipe-up dashboard; Firefox's shake-to-summarize. Design risk noted in teardowns: gestures colliding with Android's edge-back gesture — gestures must remain **redundant accelerators**, never the only path (NN/g's repeated finding that hidden navigation kills discoverability).

7. Settings organization

Chrome Android (post-2025 redesign, M3 card groups) — top-level order: 1. **You and Google** (account/sync) 2. **Basics:** Search engine, Address bar, Privacy and security, Safety check 3. **Passwords and Autofill** 4. **Advanced:** Tabs, Homepage, NTP cards, Toolbar shortcut, Notifications, Theme, Accessibility, Site settings, Languages, Downloads, About

Firefox Android — top-level order: 1. Account (sign in / sync) 2. **General:** Search, Tabs, Homepage, Customize (theme, toolbar position), Passwords, Autofill, Accessibility, Language, Set as default browser 3. **Privacy and security:** Private browsing, ETP, Site permissions, Delete browsing data, Notifications, Data collection 4. **Advanced / About**

Samsung Internet — approximate top-level order: 1. Samsung account 2. **Layout and menu** (address bar position, customize menu/toolbar) 3. Homepage / Tabs 4. Appearance 5. **Privacy** (Secret mode, Smart anti-tracking, Privacy dashboard) 6. Sites and downloads 7. Useful features

Shared skeleton: account first → high-frequency "basics" (search engine, bar position, theme) second → privacy as a named, promoted group → long-tail under "Advanced" → About last. Layout/placement settings are migrating *up* the hierarchy because bar position became a headline feature.

8. Empty states, error pages, onboarding

- **First-run flows:** Firefox's about:welcome is the documented reference — a skippable multistep flow: value proposition → set as default / pin → **auto-import from the incumbent browser** → sync sign-in. Pattern: 2–4 screens max, one ask per screen, skip always visible, sign-in never blocking.
- **Error/offline pages:** Chrome's dino game is the canonical pattern — born 2014 explicitly to "turn a frustrating moment into something fun," played by 270M+ users monthly. Structural anatomy of a good error page: friendly illustration + plain-language cause + error code (subordinate) + actionable suggestions + retry button.
- **Permission prompts:** Chrome's data showed **~85% of notification prompts were ignored or dismissed**, motivating the "quieter permission UI" (Chrome 80) and the **permission chip anchored in the omnibox** (Chrome 98) that doesn't block content. Consensus: permission requests should be non-modal, ignorable, anchored to the trust surface (address bar), and gated on user gestures.
- **Empty states:** empty tab trays get a lightweight illustration + CTA; the **private-mode landing page doubles as an explainer** (what is/isn't protected) in Chrome, Firefox, and Samsung — the mode switch itself is the empty state's job to explain, reinforced by the dark theme shift.

9. Design-system takeaways for a new browser

Evidence-backed structural rules for a new 2026 Android browser:

1. **Default the unified address bar + toolbar to the bottom; offer a top toggle.** Every major vendor now ships the option; new entrants default bottom; thumb-zone research is unambiguous.
2. **Dock, never float.** Safari's iOS 15 floating bar occluding content caused the backlash; docked-below-content is the settled pattern.
3. **Never split address bar and toolbar across screen edges.** Keep one bottom cluster.
4. **Persistent toolbar = address field + at most 4 buttons: tabs, menu, plus up to two of home/back/share — with exactly one user-configurable slot.** Full Samsung-style reordering is a later power-user feature.
5. **Tab switcher: 2-column grid of page-preview cards by default, with a list-view option,** swipe-to-close, and a dedicated private pane behind a segmented toggle.
6. **Theme-shift private mode** (dark/branded palette across bar, menus, switcher) so the mode is legible at a glance; make the private empty state the explainer.
7. **Tab count badge capped at two characters with a delight overflow (Chrome's ":D")** — constraint-as-easter-egg, zero layout cost.
8. **Menu = icon action row (back/forward/refresh/bookmark/share) + a short grouped list,** ordered: create (new tab/private) → library (history/bookmarks/downloads) → page actions (find/translate/desktop site) → settings last.
9. **Contextual actions live under long-press, never in the main menu.**
10. **NTP: search box + user-editable shortcut grid; no feed by default.** Feeds are the single largest documented complaint source across Edge, Chrome, and Firefox.

11. **Ship the standard gesture set as redundant accelerators only** — every gesture action must also have a visible control; don't collide with Android predictive back.
12. **Settings: 4 shallow groups — Account / Basics (search engine, bar position, theme) / Privacy & security / Advanced — in card containers**, frequent items ≤ 2 taps deep.
13. **Onboarding ≤ 3 skippable screens**: value pitch → import/migrate → optional default-browser ask; defer sign-in and permissions to contextual moments.
14. **Adopt the quieter-permissions chip pattern from day one**: non-modal, address-bar-anchored, ignorable prompts.
15. **Budget for one moment of delight in dead ends** (offline mini-game or animated error mascot) — Chrome's dino demonstrates outsized brand value from the cheapest surface in the product.

Desktop browser feature superset (Chrome / Firefox / Edge)

State as of mid-2026. Legend: C = Chrome, F = Firefox, E = Edge. "Mobile?" = does a mobile equivalent exist anywhere today (any browser, any platform).

1. Feature universe by category

Tab management

Feature	C	F	E	Mobile?
Tab groups (color-coded, collapsible, named)	✓	✓ (since FF137, 2025)	✓	✓ Chrome/Edge/Samsung Internet Android
Saved/synced tab groups	✓	✓	✓	✓ Chrome Android tab-group sync (2025)
AI auto-grouping / group-name suggestion	✓	✓ (FF141, on-device local model)	✓ (Copilot)	✗ rare on mobile
Tab pinning	✓	✓	✓	Partial (Samsung Internet; Chrome Android late 2025)
Vertical tabs	✗ (native experiment 2026)	✓ (FF136 sidebar)	✓ (native, pioneer)	✗ (form factor)
Sleeping/discarded tabs (auto memory reclaim)	✓ Memory Saver (ML-driven)	✓ "Unload tabs"	✓ Sleeping Tabs	✓ implicitly (Android kills tabs) but not user-controllable
Workspaces (shared/persistent named window setups)	✗	✗	✓ Edge Workspaces	✗
Tab search / fuzzy switcher	✓	✓	✓	✓ Chrome Android (2025)
Split screen / side-by-side tabs in one window	✓ (2025)	✗ (WIP)	✓	✓ Samsung Internet; OS-level split
Multi-tab selection & bulk actions	✓	✓	✓	✓ Chrome Android
Recently closed tabs / reopen	✓	✓	✓	✓
Tab preview on hover	✓	✓	✓	✗
Per-tab audio mute	✓	✓	✓	✗ mostly
Send tab to device	✓	✓	✓	✓

Bookmarks & reading

Feature	C	F	E	Mobile?
Bookmarks manager (folders, tags, search)	✓	✓ (tags, keywords — unique)	✓	✓ basic
Reading list (separate from bookmarks)	✓	✗ (Pocket killed mid-2025)	✓	✓ Chrome/Safari mobile
Collections (rich cards, notes)	✗	✗	† Killed in Edge 149 (June 2026) — cautionary tale	† was on Edge mobile
Reader mode / reading view	✓ (2026 full-page)	✓ Reader View	✓ Immersive Reader (richest)	✓ Firefox/Edge/Samsung mobile
Bookmark keyword shortcuts	✗	✓ (unique)	✗	✗

History & session

Feature	C	F	E	Mobile?
Full session restore	✓	✓ (deepest: scroll pos, form data)	✓	Partial (tabs persist)
History Journeys (topic-clustered history)	✓	✗	✓ (mobile May 2026)	✓ Edge mobile
AI natural-language history search	✓ (opt-in)	✗	✓	Rolling out with Gemini Android
Synced history & open tabs across devices	✓	✓	✓	✓

Downloads

Feature	C	F	E	Mobile?
Pause/resume downloads	✓	✓	✓	✓ Chrome Android
Parallel downloading (multi-connection)	✓ (flag)	✗	✓ (flag)	✓ Samsung Internet, third-party browsers
Download tray UX (progress ring, previews)	✓	✓	✓	Basic notification-based
Malware/dangerous-file scanning	✓ (deep scan)	✓	✓ SmartScreen	✓ partial
Scheduled downloads	✗	✗	✗	✓ Chrome Android "download later" (regional)

Privacy & security

Feature	C	F	E	Mobile?
Per-site process sandboxing / Site Isolation	✓	✓ (Fission)	✓	Partial on Android; WebView: inherited, not configurable
Safe Browsing / SmartScreen	✓ (+Enhanced real-time)	✓	✓	✓ (WebView has Safe Browsing API)
Built-in password manager + generator	✓	✓	✓	✓
Passkeys + export	✓	✓	✓	✓ (Android Credential Manager)
Password breach/leak alerts	✓	✓	✓	✓ Chrome Android
DNS-over-HTTPS (configurable)	✓	✓ (pioneer)	✓	✓ browser-level Chrome/Firefox Android; Android Private DNS is DoT system-wide
Tracker blocking by default	✗	✓ ETP Strict + Total Cookie Protection	✓ (3 tiers)	✓ Firefox/Edge/Samsung mobile
Fingerprint resistance	✗	✓	Partial	✓ Firefox Android; Brave farbling
Built-in ad blocker	✗	✗	✗ (ABP bundled on mobile)	✓ many Android browsers
HTTPS-Only / HTTPS-First	✓	✓ (pioneer)	✓	✓
One-time permissions + safety check hub	✓	✓	✓	✓ Chrome Android
VPN built-in / bundled	✗	✓ (paid, separate)	✓ Secure Network	✓ Edge mobile; Opera VPN
Scam detection (on-device AI)	✓ (Gemini Nano)	✗	✓	Rolling to Chrome Android

Profiles & identity

Feature	C	F	E	Mobile?
Multiple profiles	✓	✓ (FF144, Oct 2025)	✓ (work/personal)	✗ largely — big gap on mobile (WebView has multi-profile API since androidx.webkit 1.9!)
Guest mode	✓	✗	✓	✗
Container tabs (per-tab cookie identities)	✗	✓ Multi-Account Containers (unique)	✗	✗ nowhere on mobile — differentiation opportunity

Sync ecosystems

- All three: full sync (bookmarks, passwords, history, tabs, settings, extensions), payment/address autofill sync, cross-device handoff. Firefox is E2E-encrypted by design; Chrome offers a passphrase; Edge partial.

Developer tools

- Full DevTools suites desktop-only. Mobile: remote debugging only (WebView supports chrome://inspect). No in-browser devtools on any mobile browser — Eruda-style injected consoles are the workaround.

Performance

- Desktop: Memory Saver tiers (C), Sleeping Tabs (E), tab unloading (F); Energy saver (C/E); startup boost (C/E); preloading everywhere.
Mobile: engine-level only, no user-facing controls.

Media

Feature	C	F	E	Mobile?
Picture-in-Picture	✓	✓ (auto-PiP, subtitles — best)	✓	✓ Android OS PiP
Global media controls hub	✓	✗	✓	✓ via Android MediaSession notifications
Cast to device	✓ (Chromecast)	✗	✓ (Miracast/DLNA)	✓ Chrome Android
Live Captions	✓ (incl. translation)	✗	✓	✓ Android OS-level

Search & omnibox tricks

Feature	C	F	E	Mobile?
Site search shortcuts / custom engines	✓	✓	✓	✓ Firefox Android; Vivaldi
@-shortcuts (@bookmarks, @history, @tabs)	✓	✓ (% , * , ^ tokens)	✓	✗ mostly — open lane
Inline calculator / unit conversion	✓	✓	✓ (full converter)	✓ partially via search suggest
"Switch to tab" action chips	✓	✓	✓	✗
Quick actions ("clear history")	✓ Pedals	✓	✓	✗

Page tools

Feature	C	F	E	Mobile?
Full-page translate	✓	✓ (fully on-device — unique)	✓	✓ Chrome/Edge Android; Firefox on-device
Screenshot tool (full-page + annotate)	Partial	✓ (native full-page)	✓ Web Capture with ink	✓ Firefox Android; OS screenshot
PDF viewer with annotation	✓	✓ (edit, draw, sign — 2025)	✓ (Adobe engine)	✗ weak — most Android browsers hand off to external app (big gap)
Per-site zoom memory	✓	✓	✓	✗ mostly
Find in page (match count, highlight all)	✓	✓ (highlight all, whole-word)	✓	✓
Save page (HTML/MHTML)	✓	✓	✓	✓ Chrome offline pages; WebView saveWebArchive
QR code generator for current page	✓	✗	✓	✓ Chrome Android
PWA install & window management	✓ (richest)	✗	✓	✓ WebAPK (Chrome), Add-to-home-screen

AI features (2026 snapshot)

- Assistant sidebars everywhere (Gemini/Copilot/BYO-chatbot in Firefox); multi-tab reasoning (C/E); agentic browsing (C auto-browse, E Copilot Actions); page/PDF/video summarization universal; AI writing help (C/E); on-device small-model features (C Gemini Nano; F local models for translation/tab grouping/alt-text). Mobile is catching up fast — Gemini in Chrome Android and Copilot in Edge mobile both shipped 2026.

Accessibility

- Read aloud with natural voices (E best-in-class, C reading mode); Live captions (C/E + Android OS); reading modes with dyslexia fonts/line focus (E Immersive Reader richest); full keyboard nav/screen-reader tuning everywhere; AI alt-text (C for screen readers, F local model).

Extensions & theming

- C: Chrome Web Store (MV3 only). F: AMO, MV2+MV3 — uBlock Origin full strength. E: Edge Add-ons + CWS compat. Mobile: Firefox Android (1,000+ open ecosystem), Edge Android (curated store), Samsung (content blockers only), Chrome (none).

2. Which desktop features are migrating to mobile (trendline 2024–2026)

1. **Tab groups + tab group sync** — now table stakes on mobile.
2. **Extensions** — the biggest structural shift. Firefox Android opened its ecosystem (Dec 2023); Edge for Android testing desktop extensions; Chrome building extension support into desktop-class Android builds (Aluminium OS convergence, 2026).
3. **AI assistants** — fastest migration ever: Copilot fully in Edge mobile (May 2026); Gemini in Chrome + agentic auto browse on Android (June 2026). Desktop-first → mobile within ~12 months.
4. **Journeys / AI history** — Edge Journeys reached mobile May 2026.
5. **On-device translation** — Firefox's local-model translation shipped to Android (2025).
6. **Passkeys & breach alerts** — parity achieved via Android Credential Manager.
7. **Reading/PDF tools** — still incomplete vs desktop.
8. **Counter-trend — desktop features being killed, not migrated:** Edge Collections, Sidebar apps, and Drop all removed in Edge 149 (June 2026); Mozilla killed Pocket (2025). Lesson: curation/collection features lost to AI-centric surfaces. **Don't clone dead-ends.**
9. **Not migrating:** full DevTools, workspaces, vertical tabs, global media controls UI, Lighthouse.

3. Desktop features that would differentiate an Android browser (ranked, WebView-practical)

Ranked by differentiation-per-effort for a **WebView-based** browser. $\triangle$ = engine-limited; $\times$ = needs a full engine.

1. **Multi-profile / container identities** — almost no mobile browser has real profiles; Firefox containers exist nowhere on mobile. WebView supports this natively via `androidx.webkit` **Profile / ProfileStore API** (separate cookies, storage, service workers per profile). High wow, fully feasible. *Top pick.*
2. **Desktop-grade tab management:** named/colored saved tab groups, pinned tabs, tab search, "workspaces" (persistable window sets), auto-sleeping with per-tab memory badge — all pure browser-chrome UI, engine-independent. Feasible: tabs are your own state; "sleeping" = destroy WebView, keep state bundle + thumbnail.
3. **Real download manager** — parallel segmented downloads, pause/resume, scheduling, in-app previews. WebView only gives you `DownloadListener`; you own the downloader anyway, so you can beat Chrome Android easily. Fully feasible.
4. **Reading list + reader mode with offline caching** — inject Readability.js, store cleaned HTML for offline; add read-aloud via Android TTS. Pocket's death left a vacuum. Fully feasible.
5. **Per-site settings memory** (zoom level, desktop mode, dark mode, JS toggle, permissions) — Chrome Android barely does this. Trivial with WebView settings applied per-origin. Fully feasible.
6. **Screenshot + full-page capture + annotate** — full-page via drawing the WebView beyond viewport; annotation UI is yours. Feasible.
7. **In-page PDF experience** (view, annotate) — Android browsers punt PDFs to external apps; embedding a PDF renderer (`PdfRenderer` / `pdfium`) closes a real gap. Feasible, independent of WebView.
8. **Tracker/ad blocking** $\triangle$ — `shouldInterceptRequest` + filter lists + injected cosmetic CSS gets you a solid "Brave-lite." Limits: no true `webRequest` API, can't modify response bodies, performance cost on huge lists, and **uBlock-Origin-parity is not achievable** — full parity needs an engine. Ship the 80% version; still a differentiator vs Chrome Android.
9. **Send-to-device / cross-device sync** — you must build your own sync backend. Feasible but backend-heavy; encrypt E2E (Firefox-style) as a trust differentiator.
10. **AI layer (summarize, multi-tab compare, page Q&A)** — extract page text via `evaluateJavascript`, feed to a cloud LLM or Gemini Nano on-device via `AICore`. Assistant-tier feasible; agentic auto-browse partially feasible but fragile. Do summaries + tab-grouping AI first.
11. **Omnibox power features** — custom search keywords, @tabs/@history/@bookmarks scoping, quick actions, inline calculator. Pure UI. Cheap, delightful, no mobile browser does this well.
12. **Global media hub + PiP polish** — `MediaSession` notifications + Android PiP for fullscreen video feasible (custom `onShowCustomView` handling). Casting via Cast SDK feasible for detectable media URLs; $\triangle$ DRM streams can't be recast.
13. **Safe Browsing + breach alerts** — WebView has built-in Safe Browsing (`setSafeBrowsingEnabled`, default on) — free win; leak alerts via HIBP-style API for saved logins.

Flag: needs a full engine — do not attempt with WebView: - $\times$ **Extensions (WebExtensions API)** — no WebView support, period. Nearest substitutes: user-script injection (Via/Quetta model) and a curated "script store." Position it as "scripting," never "extensions." - $\times$ **Full DevTools UI** — remote debugging works (`setWebContentsDebuggingEnabled`); Eruda-style injected consoles possible; native DevTools = engine. - $\times$ **DNS-over-HTTPS provider choice / network-stack controls** — WebView's network stack is not configurable (system Private DNS only). - $\times$ **Site isolation tuning, fingerprint resistance (RFP), Total-Cookie-Protection semantics** — engine internals. Don't market security you can't implement. - $\times$ **HTTP-level features:** ECH toggles, response-body filtering. - $\triangle$ **On-device translation (Firefox-style)** — approximable with ML Kit Translate + DOM text-node rewriting (works, imperfect on dynamic pages).

Strategic read: WebView caps you on privacy-engineering and extensions — the two axes Firefox/Brave own. The open lanes are **organization (profiles/containers/workspaces/tab power), downloads, reading/offline, per-site control, and pragmatic AI** — all chrome-layer, all under-served on Android, all fully buildable on WebView.